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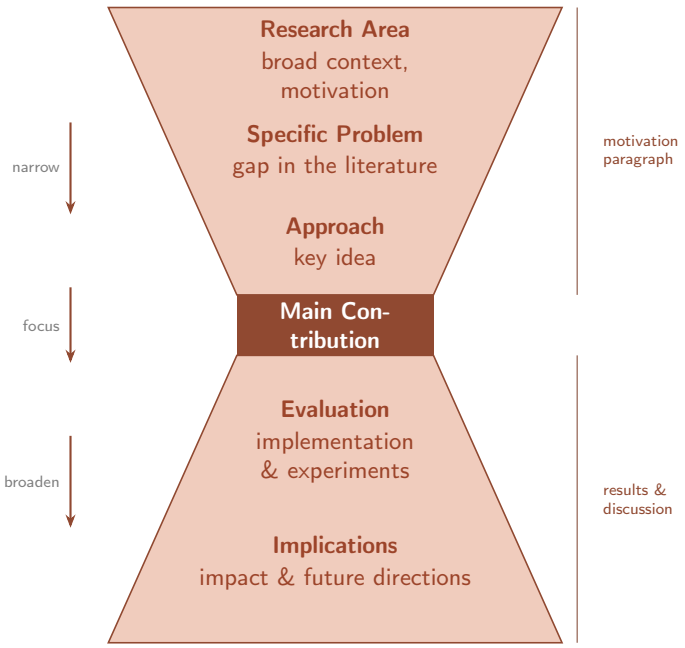
Expected Visiting Times in Interval Markov Chains

by
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Abstract



Hourglass structure of a good abstract. Start broad (research area and problem), funnel down to the specific contribution, then widen again to state results and implications.

Write a self-contained summary of **roughly 150–250 words** following the hourglass structure above. Cover the following five points in order:

1. **Research area.** Name the field (e.g., strategy synthesis, probabilistic program verification, operational semantics) and state in one sentence what the field studies and why it matters.
2. **Specific problem.** State precisely which problem this thesis addresses and why it is challenging or not yet fully solved by existing approaches.
3. **Approach.** Describe the key idea of your contribution in one or two sentences. Avoid mathematical notation and abbreviations, and use easily understandable language.
4. **Theoretical results.** Name the main property proven (soundness, completeness, an equivalence result, or a complexity bound) and state it informally.

5. **Implementation and evaluation (if applicable).** Name the tool and the benchmark set and give the headline result: how many more examples are solved compared to the baseline and what is the runtime overhead.
6. **Implications.** Explain how your results solved the issue or filled the gap you pointed out beforehand.

Acknowledgements

I want to express my most profound appreciation to my supervisor!

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Chapter 1

Introduction

Stochastic processes are powerful Stochastic processes offer effective methods for analyzing real-world systems across fields like computer science, economics, and the social sciences [GT02, BLNW14, Afr20, Gal17]. One such widely used stochastic process are *discrete-time Markov chains* (DTMCs) which were introduced by Markov in 1906 [Mar06]. DTMCs consist of a set of states and a fixed transition probability function that specifies the probability of transitioning between states. They find application in a broad range of different domains, such as computer networks and software systems [GT02, PPH13], opinion dynamics [Gal17, FJ90], and biology [RC18].

When working with Markov Chains there exist multiple measures that are of interest, like reachability probabilities, stationary distributions, and expected rewards [Kat16]. Another quantity whose importance has been recognized by many authors [MKQW25, Cas09, RC18, MAK06, Afr20, GT02, SJT05, ST06, Gok04, MMAG11, PRT10, FRG13] are *expected visiting times* (EVTs), which measure how often a given model is expected to enter each state. Similarly to reachability probabilities and expected rewards, EVTs can be computed by finding the unique solution of a linear equation system [KS76].

These EVTs are used in various areas, such as determining the expected resource demands of software systems [GT02, GWHT04, SJT05, ST06] and provide insights for the quantitative evaluation of reliability [Gok04, MMAG11, PRT10] and security of computer systems [OA14, AN14].

However, a fixed probability function can not always be assumed in practice [JL91, Kuz91, Wal96, KU02], for example, when the probability function is estimated through statistical experiments, which only provide bounds on the transition probabilities. To capture this potential imprecision of the transition probability function, *interval-valued discrete-time Markov chains* (IDTMCs) were introduce in [JL91, KU02]. These IDTMCs extend DTMCs by defining a range within which the transition probability function is assumed to lie. Various semantics have been proposed to resolve the uncertainty which arises from this imprecision. Two widely used semantics are the

once-and-for-all semantic, also known as the *Uncertain Markov Chain (UMC) semantic*, [JL91, SVA06] and the *IMDP semantic* [SVA06].

As an IDTMC characterizes a set of DTMCs, the states of an IDTMC can have multiple different EVTs belonging to the contained DTMCs. We thus focus, in this thesis, on calculating the minimum and maximum EVTs for the states of an IDTMC. If we apply the aforementioned linear equation system, that can be used to calculate EVTs in DTMCs, to IDTMCs, it thus becomes an optimization problem. Moreover, the linear equation system depends on the specific transition probability function of the given DTMC. Therefore, when extending this linear equation system IDTMCs, we have to treat the transition probability function as variable as well, due to the imprecision of the transition probability function of IDTMCs. Thus, applying the linear equation system to IDTMCs to calculate the minimum and maximum EVTs necessitates solving nonlinear optimization problems. As many nonlinear optimization problems can only be solved in nondeterministic polynomial time (NP), we propose an alternative method to calculate the minimum and maximum EVTs for IDTMCs in this thesis. To our knowledge, this has not yet been studied in the literature.

1.1 Related Work

In this section, we elaborate on related work to position this thesis in the current research starting with the origins of EVTs. We continue by elaborating on topics such as expected rewards and reachability probabilities in IDTMCs.

The concept of EVTs was introduced in 1960 by Kemeny and Snell [KS76] for *absorbing DTMCs*, which consist of *absorbing* and *transient* states. The former describe states that cannot be left after they have been entered, i.e., these states only transition back to themselves. In contrast, the latter describe states that are eventually not entered again. The authors showed that the EVTs can be determined using the so-called *fundamental matrix*. In particular, the entry in the i-th row and j-th column of the fundamental matrix equals the EVT of the transient state s_j given that the unique initial state is the transient state s_i. Importantly, Kemeny and Snell also establish the notion of EVTs that coincide with the unique solution of a linear equation system.

As aforementioned, this thesis focuses on computing the minimum and maximum EVTs for the states of IDTMCs. There are multiple measures that are related to EVTs such as *expected rewards* [MKQW25] and *reachability probabilities* [BKL$^+$17, MKQW25]. Both of these measures have been studied in IDTMCs.

As shown in [CSH08] and [SVA06], IDTMC model checking under the IMDP semantic can be reduced to model checking a Markov Decision Process (MDP). There exist many algorithms and tools to calculate expected rewards for MDPs. However, this construction results in an "exponential blowup" as [Spr18] puts it, i.e., the constructed MDP is exponential in the size of the IDTMC [CSH08].

Moreover, as shown in [BDL+17], for every IDTMC under the UMC semantic, a parametric Markov Chain can be constructed that contains the same set of DTMCs as the corresponding IDTMC. Furthermore, expected rewards for pMCs with a fixed number of parameters is in PTIME [HBK17], i.e., computing expected reward for pMCs is exponential in the number of parameters [JJK22]. However, the construction shown in [BDL+17] creates a parameter for each interval. As we cannot bound the number of intervals of the IDTMCs without loss of generality this approach is not in PTIME.

As IDTMCs are a special case of Bounded Parameter Markov Decision Processes (BMDP), a different approach is to use the results of [GLD00, TB07] for calculating expected rewards. While [GLD00] show that expected discounted rewards for BMPDs can be calculated in PTIME, [TB07] expand this result to expected rewards.

In [Teo13] presents the calculation of expected costs, i.e., expected rewards, in IDTMCs as an optization problem. However, the objective function of this optimization problem consists of a matrix inverse. As aforementioned, the computation of an inverse becomes infeasible for large models.

Furthermore, calculating minimum and maximum reachability probabilities in IDTMCs is in PTIME for both the UMC and IMDP semantic. Moreover, these two semantics coincide on reachability probabilities [BDL+17, CHK13]. To calculate the EVT of a state, we require two different reachabilty probabilities [BKL+17]. However, an optimal reachability probability is satisfied in an IDTMC if the reachability probability is fulfilled in at least one satisfying DTMC. When computing the minimum or maximum reachability probabilities for IDTMCs, we thus need to minimize or maximize these reachability probabilities

1.2 Contribution

1.3 Outline

We begin with an overview of the notation and theoretical foundations in Chapter 2. We continue, by introducing Markov Chains, related notions, and EVTs as well as relating EVTS and quantitative reachability in Chapter 3. Furthermore, we introduce the notion of Interval Markov Chains and related notions in Chapter 4.

Thereafter in Chapter 5, we define and examine EVTs in Interval Markov Chains. To this end, we start by defining EVTs for IMCs under both semantics and provide characterizations of EVTs. We then continue, by relating EVTs with quantitative reachability and showing that the two semantics coincide on EVTs in Sec. 5.1. Then, we show how EVTs can be calculated under both the UMC and IMDP semantic in Sec. 5.2.1 and Sec. 5.2.2. Lastly, we prove P-completeness for the decisions problem of EVTs in Interval Markov Chains in Sec. 5.3. Moreover, we elaborate on how stationary distributions can be calculated in Interval Markov Chains in Chapter 6. Finally, we

summarize the thesis and present possible directions for future work in Chapter 7.

Chapter 2

Preliminaries

In this chapter we introduce the theoretical foundations that are required and used throughout this thesis. We start in Sec. 2.1 by discussing the general notation used in this thesis. Rest of Overview of the sections ...

2.1 Notation

We denote the set of natural numbers, including zero, as $\mathbb{N}$ and the set of positive natural numbers, excluding zero, as $\mathbb{N}_{>0}$. The set of real numbers is denoted as $\mathbb{R}$ and the extended real numbers by $\overline{\mathbb{R}} = \mathbb{R} \cup \{-\infty, \infty\}$ for which we use the following conventions:

(i) $x + \infty = \infty + x = \infty$ for all $x \in \mathbb{R}$,

(ii) $\infty + \infty = \infty$, and

(iii) $x \cdot \infty = \infty \cdot x = \begin{cases} \infty & \text{if } x \in \mathbb{R}_{>0}, \\ 0 & \text{if } x = 0, \\ -\infty & \text{if } x \in \mathbb{R}_{<0}. \end{cases}$

The conventions are defined analogously for $-\infty$. Other expressions such as $\infty - \infty$ and $\frac{\infty}{\infty}$ are undefined. For a given set A we denote its powerset as $2^X = \{Y \mid Y \subseteq X\}$.

Furthermore, we denote the entry in the i-th row and j-th column of a matrix $\mathcal{T}$, as $(\mathcal{T})_{ij}$. We denote the i-th entry of a vector v as $v(i)$. For convenience, we use the notation $\underline{n} = \{0, 1, 2, \ldots, n\}$ with $n \in \mathbb{N}$. Moreover, for a given set S we use the notation $|S|$ to denote its cardinality.

Lastly, we use *Iverson Brackets* $[\![B]\!]$ to denote the function that maps a Boolean ex-

pression B to 1 or 0 such that

$$[\![B]\!] = \begin{cases} 1 & \text{if B is true,} \\ 0 & \text{otherwise.} \end{cases}$$

2.2 Theoretical Foundation

In this section, we introduce basic concepts from the subjects matrix, domain, measure, and probability theory which used throughout this thesis.

We start by introducing notions from domain theory. The provided definitions are taken from [MKQW25] whose material originates from [AGM95, Chapters 1].

Definition 2.1 (Partial Order)
Let X be a set with a relation $\sqsubseteq \subseteq X \times X$. The relation $\sqsubseteq$ is called a *partial order* if the following holds for all $x, y, z \in X$:

(i) $x \sqsubseteq x$ (reflexivity),

(ii) $x \sqsubseteq y$ and $y \sqsubseteq x$ implies $x = y$ (antisymmetry), and

(iii) $x \sqsubseteq y$ and $y \sqsubseteq z$ implies $x \sqsubseteq z$ (transitivity).

A partial order $\sqsubseteq$ on a set X is said to be a *linear order* if any two elements in X are comparable by $\sqsubseteq$.

We continue by briefly presenting some concepts from measure theory.

The following contents are taken from [MKQW25] whose material originates from [ADD00, Chapters 1, 4, and 5].

Definition 2.2 (σ-Algebra, Measurable Space)
Let Ω be a non-empty set. We call $\Sigma \subseteq 2^\Omega$ a *σ-algebra* on Ω if the following conditions are satisfied:

(i) $\Omega \in \Sigma$,

(ii) $X \in \Sigma$ implies $\Omega \backslash X \in \Sigma$ for all $X \subseteq \Omega$, and

(iii) $X_1, X_2, \cdots \in \Sigma$ implies $\bigcup_{i \in \mathbb{N}} X_i \in \Sigma$ for all $X_1, X_2, \cdots \subseteq \Omega$.

If Σ is a *sigma*-algebra on Ω, (Ω, Σ) is called a *measurable space*.

For the extended real numbers $\overline{\mathbb{R}}$, we consider the measurable space $(\overline{\mathbb{R}}, \mathcal{B}(\overline{\mathbb{R}}))$, where $\mathcal{B}(\overline{\mathbb{R}})$ is the *Borel σ-algebra* on the extended real numbers.

Definition 2.3 (Borel σ-Algebra on the Extended Real Numbers)
The Borel σ-algebra on $\overline{\mathbb{R}}$ is defined as the smallest σ-algebra $\mathcal{B}(\overline{\mathbb{R}})$ containing all intervals $(a, b] = \{x \in \overline{\mathbb{R}} \mid a < x \leq b\}$ with $a, b \in \overline{\mathbb{R}}$.

For a measurable space (Ω, Σ), a set $A \subseteq \Omega$ is called *measurable* if $A \in \Sigma$. The measurability of a function is defined as follows.

Definition 2.4 ((Borel) Measurable Function)
Let (Ω, Σ) and (Ω', Σ') be measurable spaces. A function $f : \Omega \to \Omega'$ is called *measurable* relative to Σ and Σ' if $\{\omega \in \Omega \mid f(\omega) \in f(E)\} \in \Omega$ for all $E \in \Sigma'$.

A function $f : \Omega \to \overline{\mathbb{R}}$ is called *Borel measurable* (on (Ω, Σ)) if it is measurable relative to Σ and $\mathcal{B}(\overline{\mathbb{R}})$.

Moreover, the class of Borel measurable functions is closed under addition.

Theorem 2.5 (Addition of Borel Measurable Functions)
Let (Ω, Σ) be a measurable space and let $f, g : \Omega \to \overline{\mathbb{R}}$ be two Borel measurable functions. The function $f + g$, defined by pointwise addition, is measurable if it is well-defined, i.e., if $f(\omega) = \infty$ implies $g(\omega) \neq -\infty$ and $f(\omega) = -\infty$ implies $g(\omega) \neq \infty$ for all $\omega \in \Omega$.

Additionally, to discuss the probability of measurable sets, we introduce the notion of *probability measures*.

Definition 2.6 (Probability Measure, Probability Space)
Let (Ω, Σ) be a measurable space. A function $\Pr : \Sigma \to [0, 1]$ is a *probability measure* on (Ω, Σ) if $\Pr(\Omega) = 1$ and for every countable collection $A_1, A_2, \ldots \in \Sigma$ of disjoint sets it holds that

$$\Pr\left(\bigcup_{k \geq 1} A_k\right) = \sum_{k \geq 1} \Pr(A_k).$$

If $\Pr : \Sigma \to [0, 1]$ is a probability measure on (Ω, Σ), then $(\Omega, \Sigma, \Pr)$ is called a *probability space*.

In a probability space $(\Omega, \Sigma, \Pr)$, the elements of Ω are called outcomes, while the measurable subsets of Ω are referred to as events. An event $E \in \Sigma$ is said to hold *almost surely* if $\Pr(E) = 1$.

In this thesis, we make use of the Lebesgue integral $\int_\Omega f(\omega) dPr(\omega)$ of a Borel measurable function $f : \Omega \to \overline{\mathbb{R}}$ and a probabiltiy measure Pr on a probability space (Ω, Σ). We presuppose that the reader is familiar with the notion of Lebesgue integration. A thorough examination can be found in [ADD00, Chapter 1.5].

Definition 2.7 (Expected Values)
Let (Ω, Σ, Pr) be a probability space. If $f : \Omega \to \overline{\mathbb{R}}$ is a Borel measurable function on (Ω, Σ, Pr), then the *expected value* of f, denoted as $\mathbb{E}[f]$, is the Lebesgue integral of f with respect to the probability measure Pr, i.e., $\mathbb{E}[f] = \int_\Omega f(\omega) dPr(\omega)$, provided the integral is well-defined.

A *probability distribution* is a function that assigns probabilities to outcomes.

Definition 2.8 (Discrete Probability Distribution)
A function P is a *discrete probability distribution* with $P : X \to [0, 1]$, where X is a countable set and $\sum_{x \in X} P(x) = 1$.

We continue by introducing notions from matrix theory based on [**?**].

An important notion, which we require in this thesis, are *(strictly) row-stochastic matrices*.

Definition 2.9 (Row-Stochastic Matrix)
A square matrix $\mathcal{T} \in \mathbb{R}_{\geq 0}^{n \times n}$ is row-stochastic, if and only if $\sum_{j=0}^{n} (\mathcal{T})_{ij} = 1$ for all rows $i \in \underline{n}$.

When considering each row as a function $f_i(j) = (\mathcal{T})_{ij}$, these functions are discrete probability distributions.

Definition 2.10 ((Strictly) Row-Substochastic Matrix)
A square matrix $\mathcal{T} \in \mathbb{R}_{\geq 0}^{n \times n}$ is row-substochastic, if and only if $\sum_{j=0}^{n} (\mathcal{T})_{ij} \leq 1$ for all rows $i \in \underline{n}$.
$\mathcal{T}$ is strictly row-substochastic if and only if ther exists at least one row i' with $\sum_{j=0}^{n} (\mathcal{T})_{i'j} < 1$

If a matrix A converges to the null matrix, then the inverse of $(\boldsymbol{I} - A)$, where $\boldsymbol{I}$ is the identity matrix, is equal to the so-called Neumann series [Neu77].

Definition 2.11 (Matrix inversion by the Neumann Series)
Given a matrix $A \in \mathbb{R}^{n \times n}$ with $\lim_{k \to \infty} A^k = 0$, i.e. A is convergent, the Neumann series $\sum_{k=0}^{\infty} A^k$ converges against $(\boldsymbol{I} - A)^{-1}$.

We will now shortly introduce linear and nonlinear programming based on [LY08].

Definition 2.12 (Linear Programming)
A *linear programming* problem with n variables and m constraints can be formu-

lated as:

$$\begin{aligned} \text{minimize} \quad & c^\mathsf{T} x \\ \text{s.t.} \quad & Ax \leq b \\ & x \geq 0 \ , \end{aligned}$$

where $c, x \in \mathbb{R}^n$, $A \in \mathbb{R}^{m \times n}$, and $b \in \mathbb{R}^m$.

In contrast to a linear programming, a nonlinear programming has a nonlinear objective function or nonlinear constraints.

Definition 2.13 (Nonlinear Programming)
A *nonlinear programming* problem with n variables and $m + r$ constraints can be formulated as:

$$\begin{aligned} \text{minimize} \quad & f(x) & & \\ \text{s.t.} \quad & g_i(x) \leq 0 & & i \in \underline{m} \\ & h_j(x) = 0 & & j \in \underline{r} \\ & x \in S \ , & & \end{aligned}$$

where $x \in \mathbb{R}^n$ and $f, g_i, h_j : S^n \to \mathbb{R}$ for all $i \in \underline{m}$, $j \in \underline{r}$. At least one of the functions f, g_i, or h_j is nonlinear.

Lastly, we use the convergence of the geometric series in this thesis.

Definition 2.14 (Convergence of Geometric Series)
The geometric series $\sum_{k=0}^{\infty} q^k$ converges for all $q \in \mathbb{R}$ with $|q| < 1$, where

$$\sum_{k=0}^{\infty} q^k = \frac{1}{1-q} \ , \quad \text{for } |q| < 1$$

Chapter 3

Markov Chains

The previous chapter provided preliminary mathematical notions, including .... In the following we introduce the notion of Markov Chains, which we require in (the main part of this) thesis.

This chapter starts with a formal definition of Markov Chains in Sec. 3.1. We then continue, by elaborating on related concepts, such as the probability space, reachability probabilities, as well as recurrent and transient states. Furthermore, we describe a further important concept, the *expected visiting times* of states in DTMCs, in Sec. 3.2 which were first introduced by [KS76].

Lastly, we relate these *expected visiting times* to reachability probabilities in Sec. 3.3. (For which we provide a short proof ...)

3.1 Discrete-Time Markov Chains

In this section, we will introduce definitions and theorems for Markov Chains. The notation and contents of this section are mostly based on [BK08], [MKQW25], and [CSH08].

Definition 3.1 (DTMC)
A discrete-time Markov chain (DTMC) is a tuple $\mathcal{D} = (S^{\mathcal{D}}, \mathbf{P}^{\mathcal{D}}, \iota_{init}^{\mathcal{D}})$, where

(i) S is a countable, nonempty set of states,

(ii) $\mathbf{P} : S \times S \to [0,1]$ is the *transition probability function*, with $\sum_{s' \in S} \mathbf{P}(s,s') = 1$ for all states $s \in S$, and

(iii) $\iota_{init} : S \to [0,1]$ is the *initial distribution*, such that $\sum_{s \in S} \iota_{init}(s) = 1$.

Whenever the DTMC $\mathcal{D}$ is evident from the context, we omit the superscript $\mathcal{D}$ from $S^{\mathcal{D}}$, $\mathbf{P}^{\mathcal{D}}$, and $\iota_{init}^{\mathcal{D}}$. We assume in this thesis, that the DTMCs are *finite*, i.e., that the

set of states S is *finite*. We call a state $s \in S$ is called *initial* if it satisfies $\iota_{init}(s) > 0$. Furthermore, we assume an arbitrary but fixed linear order $\prec$ on the set S, such that the states $s \in S$ can be indexed. Thus, $s_i \prec s_j$ if $s_i, s_j \in S$ and $i < j$. This allows us, for example, to denote the i-th entry of the initial distribution as $\iota_{init}(s_i)$, when viewing ι_{init} as a column vector. Due to the conditions (i) and (ii) provided in Def. 3.1, the transition probability function $\mathbf{P}$ and the initial distribution ι_{init} are discrete probability distributions, see Def. 2.8.

Moreover, if the probability to transition from a state s_i to s_j, for two given states $s_i, s_j \in S$, is greater than zero, i.e., $\mathbf{P}(s_i, s_j) > 0$, then we refer to s_j as a direct successor of s_i. Conversely, we refer to s_i as a direct predecessor of s_j. Additionally, we define a neighbour set N, with $s_j \in N(s_i)$, if and only if s_j is a direct successor of s_i.

Paths and Probability Space. To reason about the probabilistic behavior of a DTMC $\mathcal{D}$, we define its probability space, in which the outcomes correspond to the infinite paths of $\mathcal{D}$.

To this end, we fix a DTMC $\mathcal{D} = (S, \mathbf{P}, \iota_{init})$ for the rest of this section.

Definition 3.2 (Path of a DTMC)
An *infinite path* π in a DTMC $\mathcal{D}$ is a sequence $\pi = (s_0, s_1, \dots)$ of states with $s_0, s_1, \dots \in S$, such that $\mathbf{P}(\pi[k], \pi[k+1]) > 0$. A *finite path* is a finite prefix of an *infinite path*.

We use the notation $\pi[k] \in S$ to denote the k-th element of the path π for all $k \in \mathbb{N}_0$, i.e. $\pi[k] = s_k$.

Notice, that for any path π a given state $\pi[k]$, which is visited in step $k \in \mathbb{N}$, is always followed by one of its direct succesors, i.e., $\pi[k+1] \in N(\pi[k])$. Furthermore, we define the length $|\pi|$ of a finite path $\pi = (s_0, \dots, s_k) \in Paths^{\mathcal{D}}_{fin}$ to be equal to k, which we denote as $|\pi| = k$. As an infinite path is infinite in lenght, we define $\pi \in Paths^{\mathcal{D}}_{inf}$ to be $|\pi| = \infty$. We denote the last state of a finite path $\pi \in Paths^{\mathcal{I}}_{fin}$ as $last(\pi)$, i.e., if $|\pi| = n$, then $last(\pi) = \pi[n]$. The sets of all infinite and finite sets are denoted as $Paths^{\mathcal{D}}_{inf}$ and $Paths^{\mathcal{D}}_{fin}$, respectively.

Further, we construct the *probability space* of a DTMC as follows. The events correspond to sets of infinite paths that can be expressed as countable unions and intersections of cylinder sets, i.e., sets containing all possible continuations of finite paths in $\mathcal{D}$.

Definition 3.3 (Probability Space of a DTMC)
The *probability space* of a DTMC $\mathcal{D}$ is defined as $(Paths^{\mathcal{D}}_{inf}, \sum_{Paths^{\mathcal{D}}_{inf}}, Pr^{\mathcal{D}})$, where

- $\sum_{Paths^{\mathcal{D}}_{inf}}$ denotes the smallest σ-algebra containing the cylinder sets $Cyl(\pi')$

for all finite paths $\pi' \in Paths^{\mathcal{D}}_{fin}$ with

$$Cyl(\pi') = \{\pi \in Paths^{\mathcal{D}}_{inf} \mid \forall k \in \underline{|\pi'|} : \pi[k] = \pi'[k]\}$$

- $Pr^{\mathcal{D}}$ is the unique probability measure on $(Paths^{\mathcal{D}}_{inf}, \sum_{Paths^{\mathcal{D}}_{inf}})$ satisfying

$$Pr^{\mathcal{D}}(Cyl(\pi')) = \iota^{\mathcal{D}}_{init}(\pi'[0]) \cdot \prod_{k=0}^{|\pi'|-1} \mathbf{P}^{\mathcal{D}}(\pi'[k], \pi'[k+1])$$

for all finite paths $\pi' \in Paths^{\mathcal{D}}_{fin}$

Additionally, we use the notation $Pr^{\mathcal{D}}_s$ to denote the probability measure on the measurable space induced by fixing $s \in S$ as the unique initial state. Whenever the DTMC $\mathcal{D}$ is evident from the context, we omit the superscript $\mathcal{D}$ from $Pr^{\mathcal{D}}$ and $Pr^{\mathcal{D}}_s$.

Reachability probability. In the following, we consider events, i.e., measurable sets of infinite paths, which are relevant for the introduction of reachability probabilities in DTMCs.

First, we define the *until* operator (U), see [BK08].

Definition 3.4 (Until Operator)
Let $R, T \subseteq S^{\mathcal{D}}$ be two sets of states of a DTMC $\mathcal{D}$. For these two sets, we define the *until* operator as

$$R \,\mathsf{U}\, T = \{\pi \in Paths^{\mathcal{D}}_{inf} \mid \exists k \in \mathbb{N} : \pi[k] \in T \wedge \forall i < k : \pi[i] \in R\} \ .$$

Futhermore, the *step-bounded until* operator is defined as

$$R \,\mathsf{U}^{=k}\, T = \{\pi \in Paths^{\mathcal{D}}_{inf} \mid \pi[k] \in T \wedge \forall i < k : \pi[i] \in R\} \ .$$

Intuitively, the set $R \,\mathsf{U}\, T$ contains all paths that only visit states in R, until a state in T is visited.

Using the *until* operator, we can introduce the *eventually* operator ($\Diamond$).

Definition 3.5 (Eventually Operator)
Let $T \subseteq S$ be a set of states of a DTMC $\mathcal{D}$. Then we define the *eventually* operator as $\Diamond T = S \,\mathsf{U}\, T$.

Consequently, we define the *step-bounded eventually* operator as $\Diamond^{=k} T = S \,\mathsf{U}^{=k}\, T$.

In other words, $\Diamond T$ contains all paths $\pi \in Paths^{\mathcal{D}}_{fin}$ that visit at least one $t \in T$ at some point, i.e., there exists at least one $k \in \mathbb{N}$ with $k \leq |\pi|$, such that $\pi[k] = t$.

With the *eventually* operator, we can define the *reachability probability*.

Definition 3.6 (Reachability Probability)
The reachability probability of a set $T \subseteq S^{\mathcal{D}}$ in a DTMC $\mathcal{D}$ is given by $Pr(\Diamond T)$.

Thus, the reachability probability describes the probability of visiting a set of states in a DTMC $\mathcal{D}$. Intuitively, if $Pr(\Diamond T) > 0$ at least one state $t \in T$ is reachable in $\mathcal{D}$, i.e., there exists at least one $\pi \in Paths^{\mathcal{D}}_{fin}$, such that $\pi[0]$ is an initial state of $\mathcal{D}$ and π visits t. In case $T = \{t\}$ is a singleton set, we omit the braces and denote $\Diamond t$ instead of $\Diamond\{t\}$ (and $S \cup t$ instead of $S \cup T$). Lastly, we denote $\Diamond_t T = \{\pi \in \Diamond T \mid \pi[0] = t\}$.

To calculate the reachability probability in this thesis, we often use the recursive formula, found in the following lemma.

Lemma 3.7 (Recursive Formula for Reachability Probability)
For states $t \in S$ and $T \subseteq S$:

$$Pr_t(\Diamond T) = \sum_{u \in N(t) \cap I_{tT}} \mathbf{P}(t,u) \cdot Pr_u(\Diamond T)$$

with $Pr_t(\Diamond T) = 1$ for all $t \in T$.

We will provide a short proof for this lemma.

Proof 1
Using Def. 3.3,

$$\begin{aligned}
Pr^{\mathcal{D}}_t(\Diamond T) &= \int_{\pi \in \Diamond T} \iota^{\mathcal{D}}_{init}(\pi[0]) \cdot \prod_{k=0}^{|\pi|-1} \mathbf{P}^{\mathcal{D}}(\pi[k], \pi[k+1]) \\
&= \int_{\pi \in \Diamond_t T} \prod_{k=0}^{|\pi|-1} \mathbf{P}^{\mathcal{D}}(\pi[k], \pi[k+1]) && \text{(as t is the unique initial state, i.e., } \iota_{init}(t) = 1.\text{)} \\
&= \sum_{u \in N(t)} \int_{\pi \in \Diamond_t T \wedge \pi[1] = u} \prod_{k=0}^{|\pi|-1} \mathbf{P}^{\mathcal{D}}(\pi[k], \pi[k+1]) && \text{(as } t \text{ can only be succeded by } u \in N(t)\text{)} \\
&= \sum_{u \in N(t)} \mathbf{P}(t,u) \int_{\pi \in \Diamond_t T \wedge \pi[1] = u} \prod_{k=1}^{|\pi|-1} \mathbf{P}^{\mathcal{D}}(\pi[k], \pi[k+1]) && \text{(with } \pi[0] = t, \pi[1] = u\text{)}
\end{aligned}$$

For $\pi \in \Diamond_t T$ with $\pi[1] = u$, define $\pi' = (\pi[1], \dots, last(\pi))$. As $\pi'[0] = u$ and $last(\pi') = s$, $\pi' \in \Diamond_u T$. Conversely, for $\pi' \in \Diamond_u T$, define $\pi = (t, \pi'[0], \dots, last(\pi'))$, then $\pi \in \Diamond_t T$. Thus,

$$
\begin{aligned}
&= \sum_{u \in N(t)} \mathbf{P}^{\mathcal{D}}(t,u) \cdot Pr_u^{\mathcal{D}}(\Diamond T) \\
&= \sum_{u \in N(t) \cap I_{tT}} \mathbf{P}^{\mathcal{D}}(t,u) \cdot Pr_u^{\mathcal{D}}(\Diamond T) && \text{(as, if } u \notin I_{tT}\text{, then} \\
& && Pr_u(\Diamond T) = 0\text{, see Def. 5.7.)} \\
& && (3.1)
\end{aligned}
$$

□

Lastly, we use the *next* operator ($\bigcirc$), which is defined as follows.

Definition 3.8 (Next Operator)
Let $\mathcal{D}$ be a DTMC and let $E \in \sum_{Paths_{inf}^{\mathcal{D}}}$ be a measurable set of infinite paths in $\mathcal{D}$. We now define the *next* operator, as

$$\bigcirc E = \{\pi \in Paths_{inf}^{\mathcal{D}} \mid (\pi[1], \pi[2], \dots) \in E\} \ ,$$

In other words, a path $\pi \in Paths_{inf}^{\mathcal{D}}$ is contained in $\bigcirc E$ if and only if $\pi' = (\pi[1], \pi[2], \dots)$ is contained in E. Moreover, the next step probability is defined as $Pr(\bigcirc E) = \sum_{s' \in N(s)} \mathbf{P}(s,s') \cdot Pr_{s'}(E)$.

In this thesis, we use the *next* operator ($\bigcirc$) in combination with the *eventually* operator ($\Diamond$). Specifically, we use them to denote the *recurrence probability* $Pr_s(\bigcirc\Diamond s)$, i.e., the probability to reach s from all of its successors.

Strongly Connected Components. We can partition the state set of a DTMC as shown in the following definition.

Definition 3.9 ((Bottom) Strongly Connected Component)
A non-empty set of states $C \subseteq S^{\mathcal{D}}$ of a DTMC $\mathcal{D}$ is *strongly connected* if and only if for every pair of states $s, t \in S$, there exists a path $\pi \in Paths_{inf}^{\mathcal{D}}$, such that $\pi[0] = s$ and $\pi[k]$ for some $k \in \mathbb{N}$. If C is strongly connected and no proper subset of C, i.e. $C' \subset C$, is strongly connected, then C is called a *strongly connected component* (SCC).

Furthermore, an SCC C is called a *bottom strongly connected component* (BSCC) if there exists no $k \in \mathbb{N}$ and no path $(\pi[0], \dots, \pi[k], \dots) \in Paths_{inf}^{\mathcal{D}}$ such that $\pi[0] \in C$ and $\pi[k] \notin C$.

In other words, once a state of a BSCC C is entered, C cannot be left again.

Using BSCCs, we can define *acyclic* DTMCs.

Definition 3.10 (Acyclic DTMC)
A DTMC $\mathcal{D}$ is called *acyclic* if and only if each SCC is either a BSSC or a singleton set $\{s\}$ with $\mathbf{P}(s,s) = 0$

Recurrent and Transient States. Using the *recurrence probability* $Pr_s^{\mathcal{D}}(\bigcirc\Diamond s)$, we can partition the state set into *recurrent* and *transient* states.

Definition 3.11 (Recurrent and Transient States)
For a given DTMC $\mathcal{D}$ and state $s \in S$, s is called *recurrent* if and only if $Pr_s(\bigcirc\Diamond s) = 1$. All states that are not *recurrent* are called *transient*.

In other words, a state is *recurrent* precisely, when we are guaranteed to reach it from all of its direct successors, meaning once we reach a *recurrent* state, we are bound to return to it. Using this definition, we can split the states S of a DTMC into two sets with S_{tr} containing all of the transient states and S_{re} containing all of the recurrent states. Notice, that *recurrent* states are precisely the states contained in a BSCC. For convenience, we additionally define the set of *initial* states $S_{in}^{\mathcal{D}} = \{s \mid \iota_{init}^{\mathcal{D}}(s) > 0\}$. Furthermore, we define the set of *transient initial* states $S_{tr\cap in} \subseteq S_{in}$ with $S_{tr\cap in} = S_{in} \cap S_{tr}$.

Moreover, the event $\Diamond S_{re}$, i.e., to reach any recurrent state, holds *almost surely* in any DTMC.

Theorem 3.12 (Limit Behaviour)
For any DTMC $\mathcal{D}$, set of recurrent states S_{re}, and any state $s \in S$, it holds that $\lim_{k\to\infty} Pr_s(\Diamond^{=k} S_{re}) = 1$.

Intuitively, the probability of reaching a recurrent state after k steps, starting in s, tends towards 1 as k tends towards infinity.

Lastly, we define *absorbing* states.

Definition 3.13 (Absorbing state)
A states s of a DTMC $\mathcal{D}$ is refered to as an *absorbing state* if and only if $\mathbf{P}^{\mathcal{D}}(s,s) = 1$.

This means, once the model enters an *absorbing* state, it cannot leave it. Notice, that *absorbing* states are a special type of *recurrent* states, as the only direct successor of an *absorbing* state $s \in S$ is s itself and therefore trivially $Pr_s(\Diamond s) = 1$. Thus, we are guaranteed to return to a given *absorbing* state after one step.

Definition 3.14 (Markov Descision Process (MDP))
[BK08] A *Markov Decision Process* (MDP) is a tuple $\mathcal{M} = (S, Act, A, \mathbf{P}, \iota_{init})$, where

(i) S is a countable set of states

(ii) Act is a set of actions

(iii) $A(s) \subseteq Act$ denotes the applicable actions for all states $s \in S$

(iv) $\mathbf{P} : S \times Act \times S \to [0,1]$ is the transition probability function such that $\sum_{s' \in S} \mathbf{P}(s, \alpha, s') = 1$ for all $s \in S$ and $\alpha \in A(s)$

(v) $\iota_{init} : S \to [0,1]$ is the initial distribution such that $\sum_{s \in S} \iota_{init}(s) = 1$

Definition 3.15 (Canonical Form)
For a DTMC $\mathcal{D}$, its transition matrix $\mathcal{T}$ is said to be in canonical form if and only if

$$\mathcal{T} = \begin{bmatrix} Q & R \\ 0 & S \end{bmatrix}$$

where

(i) $Q = (\mathbf{P}(s,t))_{s,t \in S_{tr}}$ contains the transition probabilities between the transient states

(ii) $R = (\mathbf{P}(s,t))_{s \in S_{tr}, t \in S_{re}}$ contains the transition probabilities between transient and recurrent state pairs

(iii) $S = (\mathbf{P}(s,t))_{s,t \in S_{re}}$ contains the transition probabilities between the recurrent states

For absorbing DTMCs $S = \boldsymbol{I}$. As mentioned in **??**, this is precisely possible for

3.2 Expected Visiting Times for DTMCs

In this section, we will introduce the concept of *expected visiting times*, which were first introduced by Kemeny and Snell [KS76] in the context of the *fundamental matrix*. The notation and contents of this section are mostly based on [MKQW25] and [KS76].

Intuitively, the notion of EVTs describes how often we expect the model to be in a given state. To formally introduce EVTs, we fix a DTMC $\mathcal{D} = (S, \mathbf{P}, \iota_{init})$ and start by introducing *visiting times* along paths.

Definition 3.16 (Visiting Times of Infinite Paths)
The function $vt_s : Paths_{inf}^{\mathcal{D}} \to \mathbb{R}$ describes the *visiting time* of a state $s \in S$ along infinite paths in $\mathcal{D}$, where for $\pi \in Paths_{inf}^{\mathcal{D}}$

$$vt_s(\pi) = \sum_{k=0}^{\infty} [\![\pi[k] = s]\!] \ .$$

Similarly, the function $vt_T : Paths^{\mathcal{D}}_{inf} \to \mathbb{R}$ descsribes the *visiting time* of a set of states $T \subseteq S$ along infinite paths in $\mathcal{D}$, where for $\pi \in Paths^{\mathcal{D}}_{inf}$

$$vt_T(\pi) = \sum_{k=0}^{\infty} [\![\pi[k] \in T]\!] \ .$$

As proven in [MKQW25], the Lebesgue integral of the non-negative function vt_s is well-defined for all states $s \in S^{\mathcal{D}}$. Using this result, we can shortly prove the same for vt_T.

Proof 2
Using def. 3.16,

$$\begin{aligned} vt_T(\pi) &= \sum_{k=0}^{\infty} [\![\pi[k] \in T]\!] \\ &= \sum_{k=0}^{\infty} \sum_{t \in T} [\![\pi[k] = t]\!] \\ &= \sum_{t \in T} vt_t(\pi) \ . \end{aligned}$$

Thus, the Lebesgue integral of the non-negative function vt_T is well-defined for all sets of states $T \subseteq S$. □

With this function, we can now define the EVT of a state s as the expected value of vt_s over all possible paths.

Definition 3.17 (Expected Visiting Times in DTMCs)
For a state $s \in S$, the *expected visiting time* of s is characterized as

$$\mathbb{E}^{\mathcal{D}}[vt_s] = \int_{\pi \in Paths^{\mathcal{D}}_{inf}} vt_s(\pi) dPr^{\mathcal{D}}(\pi) \ .$$

Additionally, we denote the *expected visiting time* that is obtained by fixing a certain state $t \in S$ as the only initial state as $\mathbb{E}^{\mathcal{D}}_t[vt_s]$. It is defined as

$$\mathbb{E}^{\mathcal{D}}_t[vt_s] = \int_{\pi \in Paths^{\mathcal{D}}_{inf}} vt_s(\pi) dPr^{\mathcal{D}}_t(\pi) \ .$$

Lastly, for a set of states $T \subseteq S$ the *expected visiting time* of T is characterized as

$$\mathbb{E}^{\mathcal{D}}[vt_T] = \sum_{t \in T} \mathbb{E}[vt_t] \ .$$

Because the probability measure of paths π, see definition Def. 3.3, is defined as a product containing the factor $\iota_{init}(\pi[0])$, it is nonzero only for paths that begin in a state s with $\iota_{init}(s) > 0$. Thus, the value $\mathbb{E}^{\mathcal{D}}[vt_s]$ gives the number of times the state s is expected to be visited from all paths in $\mathcal{D}$ that start in one of the initial states. Whenever the DTMC $\mathcal{D}$ is evident from the context, we will omit the superscript $\mathcal{D}$ from $\mathbb{E}^{\mathcal{D}}[vt_s]$, $\mathbb{E}^{\mathcal{D}}_t[vt_s]$, and $\mathbb{E}^{\mathcal{D}}[vt_T]$.

We shortly prove the third equation of Def. 3.17.

Proof 3

$$\begin{aligned}
\mathbb{E}[vt_T] &= \int_{\pi \in Paths^{\mathcal{D}}_{inf}} vt_T(\pi) dPr^{\mathcal{D}}(\pi) \\
&= \int_{\pi \in Paths^{\mathcal{D}}_{inf}} \sum_{t \in T} vt_t(\pi) dPr^{\mathcal{D}}(\pi) && \text{(using Def. 3.16)} \\
&= \sum_{t \in T} \int_{\pi \in Paths^{\mathcal{D}}_{inf}} vt_t(\pi) dPr^{\mathcal{D}}(\pi) \\
&= \sum_{t \in T} \mathbb{E}[vt_t] \ . && \text{(using Def. 3.17)}
\end{aligned}$$

□

Using the aforementioned partition of the set of states S into transient and recurrent states, see Def. 3.11, we can already give a characterization for the latter.

Lemma 3.18 (EVTs of Recurrent States)
For all recurrent states $s \in S_{re}$, it holds that

$$\mathbb{E}[vt_s] = \begin{cases} \infty & \text{if s is reachable,} \\ 0 & \text{else.} \end{cases}$$

For a proof of this lemma, see [MKQW25].

Thus, we will focus on finding the non-trivial EVTs of the transient states. For this, we will first define the *fundamental matrix* which was introduced by Kemeny and Snell [KS76].

Definition 3.19 (Fundamental Matrix)
For a DTMC $\mathcal{D}$ with the matrix $Q = (\mathbf{P}^{\mathcal{D}}(s,t))_{s,t \in S_{tr}}$ containing the transition probabilities between the transient states (defined in Def. 3.15), the *fundamental matrix* F is defined as,

$$F = (I - Q)^{-1} \ .$$

Kemeny and Snell show in [KS76] that Q converges to the null matrix. Therefore, the Neumann series of Q converges with $\sum_{k=0}^{\infty} Q^k = (\boldsymbol{I} - Q)^{-1}$, see Def. 2.11, i.e., the fundamental matrix is well-defined.

Importantly for this thesis, Kemeny and Snell show in [KS76], that the fundamental matrix gives for any two transient states $s, t \in S_{tr}$ the expected number of times s is visited, when starting in t. To be more precise, $\mathbb{E}_t[vt_s] = (F)_{ts}$. From this equation, it intuitively follows that transient states are only visited a finite amount of times.

Thus, we can use the fundamental matrix to characterize the EVTs for transient states. However, as the calculation of the inverse of a high-dimensional matrix becomes infeasible for large models, we will take a look at how the EVTs can be calculated using linear equation systems, as shown in [MKQW25].

First, we define

$$\tau_{init} = (\iota_{init}(s_1), \ldots, \iota_{init}(s_n))^{\mathsf{T}}$$

for all $s \in S_{tr}$ with which we can then introduce the following theorem taken from [MKQW25].

Theorem 3.20 (Linear Equation System Characterization of EVTs of Transient States)
The vector $(\mathbb{E}[vt_s])_{s \in S_{tr}}$ provides the unique solution of the following linear equation system in $\mathbb{R}^{|S_{tr}|}$:

$$(\boldsymbol{I} - Q^T) \cdot x = \tau_{init} \ .$$

For a proof of this theorem, see [MKQW25].

This equation system can then be rewritten as

$$x = \tau_{init} + Q^{\mathsf{T}} \cdot x \ .$$

For a single state $s \in S_{tr}$, we can thus calculate the respective EVT as follows,

$$\mathbb{E}[vt_s] = \iota_{init}(s) + \sum_{t \in S_{tr}} \mathbf{P}(t, s) \cdot \mathbb{E}[vt_t] \ . \tag{3.2}$$

3.3 Relating Expected Visiting Times and Quantitative Reachability

We will now relate *expected visiting times* to *quantitave reachability* for DTMCs.

Lemma 3.21
Given a DTMC $\mathcal{D}$ and its fundamental matrix F, we can define

$$(F)_{ts} = Pr_t(\Diamond s) \cdot (F)_{ss} \ ,$$

for all $s \neq t \in S_{tr}$, and thus

$$\mathbb{E}[vt_s] = Pr(\Diamond s) \cdot (F)_{ss} \tag{3.3}$$
$$= Pr(\Diamond s) \cdot \mathbb{E}_s[vt_s] \ . \tag{3.4}$$

Trivially, all paths that reach s from t at least once visit s. Thus, intuitively, the *expected visiting time* of s starting in t is simply the product of the probability of reaching s and the *expected visiting time* of s starting in s.

Proof 4

$$\begin{aligned}
\mathbb{E}^{\mathcal{D}}[vt_s] &= \int_{\pi \in Paths^{\mathcal{D}}_{inf}} vt_s(\pi) dPr^{\mathcal{D}}(\pi) \\
&= \int_{\pi \in Paths^{\mathcal{D}}_{inf} \wedge \exists k : \pi[k] = s} \sum_{k=0}^{\infty} [\![\pi[k] = s]\!] dPr^{\mathcal{D}}(\pi) \\
&+ \underbrace{\int_{\pi \in Paths^{\mathcal{D}}_{inf} \wedge \nexists k : \pi[k] = s} \sum_{k=0}^{\infty} [\![\pi[k] = s]\!] dPr^{\mathcal{D}}(\pi)}_{=0} \\
&= \int_{\pi \in Paths^{\mathcal{D}}_{inf} \wedge \exists k : \pi[k] = s} \sum_{k=0}^{\infty} [\![\pi[k] = s]\!] dPr^{\mathcal{D}}(\pi) \\
&= Pr(\Diamond s) \cdot \int_{\pi \in Paths^{\mathcal{D}}_{inf}} \sum_{k=0}^{\infty} [\![\pi[k] = s]\!] dPr^{\mathcal{D}}_s(\pi) \qquad (3.5)
\end{aligned}$$

□

???

Lemma 3.22
Given a DTMC $\mathcal{D}$ and its fundamental matrix F, we can define

$$(F)_{ss} = \frac{1}{1 - Pr_s(\bigcirc \Diamond s)}$$

for all $s \in S_{tr}$, where $Pr_s(\bigcirc \Diamond s) = \sum_{t \in N(s)} \mathbf{P}(s,t) \cdot Pr_t(\Diamond s)$

(The fraction is well-defined as $Pr_s(\bigcirc \Diamond s) < 1$, see Def. 3.11)

Proof 5
As the only initial state is s, it is necessarily visited at least once. After that the probability that s will be visited at least one more time is $Pr_s(\bigcirc \Diamond s)$ and the probability that it will not, is accordingly $(1 - Pr_s(\bigcirc \Diamond s))$. Moreover, the

probability of visiting s exactly k times is precisely the probability of returning $k-1$ times and not return again afterwards. Thus, $Pr_s(N_s = k) = Pr_s(\bigcirc\Diamond s)^{k-1}(1 - Pr_s(\bigcirc\Diamond s))$ for $k \in \mathbb{N}$, where N_s is a random variable representing the total number of time s is visited (Pr_s means starting in s). Intuitively,

$$
\begin{aligned}
(F)_{ss} &= \sum_{k=1}^{\infty} k \cdot Pr_s(N_s = k) \\
&= \sum_{k=1}^{\infty} k \cdot Pr_s(\bigcirc\Diamond s)^{k-1}(1 - Pr_s(\bigcirc\Diamond s)) \\
&= (1 - Pr_s(\bigcirc\Diamond s)) \sum_{k=1}^{\infty} k \cdot Pr_s(\bigcirc\Diamond s)^{k-1} \\
&= (1 - Pr_s(\bigcirc\Diamond s)) \cdot \frac{1}{(1 - Pr_s(\bigcirc\Diamond s))^2} = \frac{1}{1 - Pr_s(\bigcirc\Diamond s)} \qquad \text{(by geometric series.)}
\end{aligned}
$$

□

Finally, using **??**,

$$
\begin{aligned}
\mathbb{E}[vt_s] &= Pr(\Diamond s) \cdot (F)_{ss} && \text{(by Lemma 3.21)} \\
&= \frac{Pr(\Diamond s)}{1 - Pr_s(\bigcirc\Diamond s)} && \text{(by Lemma 3.22)} \qquad (3.6) \\
&= \sum_{t \in S_{tr \cap in}} \frac{Pr_t(\Diamond s)}{1 - Pr_s(\bigcirc\Diamond s)} \cdot \iota_{init}(t)
\end{aligned}
$$

Notice, that for any given DTMC $\mathcal{D}$ we can construct $\mathcal{D}^s_{uni} = (S^{\mathcal{D}} \cup \{s_{in}, s_{suc}\}, \mathbf{P}^{\mathcal{D}^s_{uni}}, \iota^{\mathcal{D}^s_{uni}}_{init})$, where $\mathbf{P}^{\mathcal{D}^s_{uni}}(s_i, s_j) = \mathbf{P}^{\mathcal{D}}(s_i, s_j)$, $\mathbf{P}^{\mathcal{D}^s_{uni}}(s, s_j) = 0$, $\mathbf{P}^{\mathcal{D}^s_{uni}}(s, s_{suc}) = 1$, $\mathbf{P}^{\mathcal{D}^s_{uni}}(s_{suc}, s_j) = \mathbf{P}^{\mathcal{D}}(s, s_j)$, $\mathbf{P}^{\mathcal{D}^s_{uni}}(s_{in}, s_j) = \iota^{\mathcal{D}}_{init}(s_j)$ for all $s_i \in S^{\mathcal{D}} \backslash \{s\}$ and $s_j \in S^{\mathcal{D}}$, and $\iota^{\mathcal{D}^s_{uni}}_{init}(s_{in}) = 1$. Essentially, $\mathcal{D}^s_{uni}$ is a copy of $\mathcal{D}$, where instead of using ι_{init} as an initial distribution, we simply have a single initial state s_{in} and s has only one direct successor s_{suc}, which has the same probability distribution as s in $\mathcal{D}$.

Theorem 3.23
The reachability probabilities and thus also EVTs in $\mathcal{D}$ and the corresponding $\mathcal{D}^s_{uni}$ are identical.

Proof 6
First, we show that the reachability probabilities are equal (see Def. 3.4) $Pr^{\mathcal{D}}(\Diamond T) = Pr^{\mathcal{D}^s_{uni}}(\Diamond T)$ for all $T \in S^{\mathcal{D}}$. For this, we define the bijection $\phi : Paths^{\mathcal{D}^s_{uni}}_{inf} \to Paths^{\mathcal{D}}_{inf}, \pi = (s_{in}, s_1, s_2, \dots, s_{k-1}, s_{suc}, s_k, \dots) \mapsto (s_1, s_2, \dots, s_{k-1}, s_k, \dots)$, i.e.,

the bijection ϕ removes s_{in} and all s_{suc}. This bijection is well defined, as $N(s_{in}) = \{s \mid \iota^{\mathcal{D}}_{init}(s) > 0\}$, $\mathbf{P}^{\mathcal{D}^s_{uni}}(s_i, s_{suc}) > 0$ only for $s_i = s$, and $\mathbf{P}^{\mathcal{D}^s_{uni}}(s_{suc}, s_i) = \mathbf{P}^{\mathcal{D}}(s, s_i)$ ($\rightarrow$ thus same (amount of) paths).

Moreover, assume a given $\pi \in Paths^{\mathcal{D}_{uni}}_{inf}$ has $\pi[k_i] = s_{suc}$ with $i \in \underline{j}$ for some $j \in \mathbb{N}_{>0}$ and $\phi(\pi) = \hat{\pi}$.

$$
\begin{aligned}
Pr^{\mathcal{D}_{uni}}(\pi) &= \underbrace{\iota^{\mathcal{D}_{uni}}_{init}(\pi[0])}_{=1} \cdot \prod_{k=0}^{|\pi|-1} \mathbf{P}^{\mathcal{D}_{uni}}(\pi[k], \pi[k+1]) \\
&= \mathbf{P}^{\mathcal{D}_{uni}}(\pi[0], \pi[1]) \cdot \underbrace{\prod_{i=1}^{j-1} \mathbf{P}^{\mathcal{D}_{uni}}(\pi[\underbrace{k_i - 1}_{=s}], \pi[\underbrace{k_i}_{=s_{suc}}])}_{=1} \cdot \prod_{k=1}^{k_1-2} \mathbf{P}^{\mathcal{D}_{uni}}(\pi[k], \pi[k+1]) \\
&\quad \cdot \prod_{k=k_1}^{k_2-2} \mathbf{P}^{\mathcal{D}_{uni}}(\pi[k], \pi[k+1]) \cdot \ldots \cdot \prod_{k_j}^{|\pi|-1} \mathbf{P}^{\mathcal{D}_{uni}}(\pi[k = k_j], \pi[k+1]) \\
&= \iota^{\mathcal{D}}_{init}(\phi(\pi)[1]) \cdot \prod_{k=1}^{|\hat{\pi}|-1} \mathbf{P}^{\mathcal{D}}(\phi(\pi)[k], \phi(\pi)[k+1]) \qquad (\text{as, } \pi[k_i + 1] = \hat{\pi}[k_i]) \\
&= \iota^{\mathcal{D}}_{init}(\hat{\pi}[0]) \cdot \prod_{k=0}^{|\hat{\pi}|-1} \mathbf{P}^{\mathcal{D}}(\hat{\pi}[k], \hat{\pi}[k+1]) \\
&= Pr^{\mathcal{D}}(\hat{\pi}) \ .
\end{aligned}
$$

Therefore, we can conclude $Pr^{\mathcal{D}_{uni}}(\Diamond T) = Pr^{\mathcal{D}}(\Diamond T)$, as $\Diamond T \subseteq Paths^{\mathcal{D}_{uni}}_{inf}$. Using Eq. (3.6), we can then also conclude that the EVTs of $\mathcal{D}$ and $\mathcal{D}_{uni}$ must be equal.□

Thus, the reachability probabilities $Pr^{\mathcal{D}_{uni}}_{s_{in}}(\Diamond s) = \sum_{t \in S_{in}} Pr^{\mathcal{D}}_t(\Diamond s) = \sum_{t \in S_{tr \cap in}} Pr^{\mathcal{D}}_t(\Diamond s)$ are equal and we can simplify the previous expression,

$$
\mathbb{E}^{\mathcal{D}}[vt_s] = \frac{Pr^{\mathcal{D}_{uni}}_{s_{in}}(\Diamond s)}{1 - Pr^{\mathcal{D}_{uni}}_{s_{suc}}(\Diamond s)} \cdot \tau_{init}(s_{in}) \left(= \frac{Pr^{\mathcal{D}}(\Diamond s)}{1 - Pr^{\mathcal{D}}_s(\bigcirc \Diamond s)} \cdot \tau_{init}(s_{in}) \right) \ . \tag{3.7}
$$

In other words, we can calculate the EVT of a DTMC $\mathcal{D}$ with only two reachability probabilities using the corresponding DTMC $\mathcal{D}_{uni}$.

Moreover, using Lemma 3.21 and the construction $\mathcal{D}_{uni}$, we can calculate the quantitative reachability of a given state with $Pr(\Diamond s) = \frac{(F^{\mathcal{D}_{uni}})_{s_{in}s}}{(F)_{ss}}$.

Chapter 4

Interval Markov Chains

The focus of this thesis are expected visiting times in Interval Markov Chains, which we will introduce in this chapter. In the following sections we will mainly discuss how the expected visiting times can be calculated and shortly examine stationary distributions of Interval Markov Chains.

We start by introducing concepts which we will use in the following sections.

Definition 4.1 (Interval Markov Chain)
An interval discrete-time Markov chain (IDTMC) is a tuple $\mathcal{I} = (S^{\mathcal{I}}, \underline{\mathbf{P}}^{\mathcal{I}}, \overline{\mathbf{P}}^{\mathcal{I}}, \underline{\iota}^{\mathcal{I}}_{init}, \overline{\iota}^{\mathcal{I}}_{init})$, where

(i) $S^{\mathcal{I}}$ is a countable, nonempty set of states,

(ii) $\underline{\mathbf{P}}^{\mathcal{I}} : S^{\mathcal{I}} \times S^{\mathcal{I}} \to [0,1]$ is the lower bound on the transition probability function, such that $\underline{\mathbf{P}}^{\mathcal{I}}(s_i, s_j) \leq \mathbf{P}(s_i, s_j)$ for all $s_i, s_j \in S^{\mathcal{I}}$,

(iii) $\overline{\mathbf{P}}^{\mathcal{I}} : S^{\mathcal{I}} \times S^{\mathcal{I}} \to [0,1]$ is the upper bound on the transition probability function, such that $\mathbf{P}(s_i, s_j) \leq \overline{\mathbf{P}}^{\mathcal{I}}(s_i, s_j)$ for all $s_i, s_j \in S^{\mathcal{I}}$,

(iv) $\underline{\iota}^{\mathcal{I}}_{init} : S^{\mathcal{I}} \to [0,1]$ is the lower bound on the initial distribution, such that $\underline{\iota}^{\mathcal{I}}_{init}(s) \leq \iota_{init}(s)$ for all $s \in S^{\mathcal{I}}$, and

(v) $\overline{\iota}^{\mathcal{I}}_{init} : S \to [0,1]$ is the upper bound on the initial distribution, such that $\iota_{init}(s) \leq \overline{\iota}^{\mathcal{I}}_{init}(s)$ for all $s \in S^{\mathcal{I}}$.

Whenever the IDTMC $\mathcal{I}$ is evident from the context, we omit the superscript $\mathcal{I}$ from $S^{\mathcal{I}}$, $\mathbf{P}^{\mathcal{I}}$, and $\iota^{\mathcal{I}}_{init}$. In contrast to the precise transition probability function and initial distribution of DTMCs, Interval Markov Chains allow for uncertainty.

Example 4.2
The Fig. 4.1 depicts an exemplary IDTMC $\mathcal{I}_E = (S, \underline{\mathbf{P}}, \overline{\mathbf{P}}, \underline{\iota}_{init}, \overline{\iota}_{init})$, where $S = \{s_1, s_2, s_3, s_4, s_5\}$, the lower and upper transition probability functions are given by

$$\underline{\mathbf{P}} = \begin{bmatrix} 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0.2 & 0.2 \\ 0.2 & 0 & 0 & 0 & 0.2 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix} \qquad \overline{\mathbf{P}} = \begin{bmatrix} 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0.8 & 0.8 \\ 0.8 & 0 & 0 & 0 & 0.8 \\ 0 & 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix},$$

and the lower and upper initial distributions

$$\underline{\iota}_{init} = (0.5, 0, 0, 0, 0)^\mathsf{T} \qquad \overline{\iota}_{init} = (1, 0, 0.5, 0, 0)^\mathsf{T}.$$

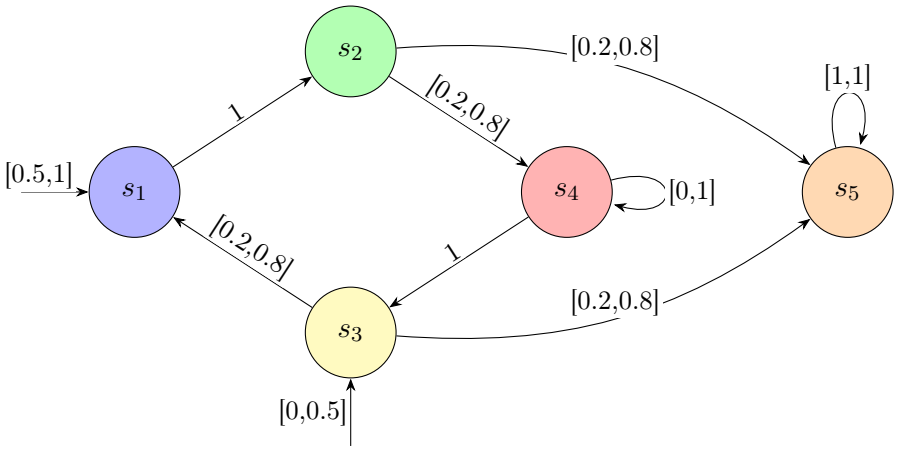


Figure 4.1: IDTMC $\mathcal{I}$

Definition 4.3 (Path of an IDTMC)
Similarly to DTMCs, an *infinite path* in an IDTMC $\mathcal{I} = (S,)$ is a sequence $\pi = (s_0, s_1, \dots)$ of states with $s_0, s_1, \dots \in S^{\mathcal{I}}$, such that $\underline{\mathbf{P}}(\pi[k], \pi[k+1]) > 0$ for all $k \in \mathbb{N}$.

Fixed-initial-state definition. In the literature, see, e.g., [CHK13], it is common to represent the initial distribution by an additional initial state rather than introducing a separate initial-distribution function. In this case, an IMC is given as a tuple $\mathcal{I}_{\text{uni}} = (S, \underline{\mathbf{P}}, \overline{\mathbf{P}}, s_{\text{in}})$, where s_{in} is the unique initial state. We refer to this as the *fixed-initial-state definition*.

More precisely, let $\mathcal{I} = (S^{\mathcal{I}}, \underline{\mathbf{P}}^{\mathcal{I}}, \overline{\mathbf{P}}^{\mathcal{I}}, \underline{\iota}^{\mathcal{I}}_{init}, \overline{\iota}^{\mathcal{I}}_{init})$ be an IMC in our definition, and let $\iota^{\mathcal{I}}_{\text{init}}$ denote its initial distribution. The corresponding fixed-initial-state representation is obtained by introducing a fresh state s_{in} and defining $S^{\mathcal{I}_{\text{uni}}} = S^{\mathcal{I}} \cup \{s_{\text{in}}\}$.

The lower and upper transition probability functions are given by

$$\underline{\mathbf{P}}^{\mathcal{I}_{\text{uni}}}(s_i, s_j) = \begin{cases} \underline{\mathbf{P}}^{\mathcal{I}}(s_i, s_j), & s_i, s_j \neq s_{\text{in}}, \\ \iota^{\mathcal{I}}_{\text{init}}(s_j), & s_i = s_{\text{in}}, \\ 0, & \text{otherwise}, \end{cases}$$

and

$$\overline{\mathbf{P}}^{\mathcal{I}_{\text{uni}}}(s_i, s_j) = \begin{cases} \overline{\mathbf{P}}^{\mathcal{I}}(s_i, s_j), & s_i, s_j \neq s_{\text{in}}, \\ \iota^{\mathcal{I}}_{\text{init}}(s_j), & s_i = s_{\text{in}}, \\ 0, & \text{otherwise}. \end{cases}$$

The same construction applies to DTMCs. Let $\mathcal{D} = (S^{\mathcal{D}}, \mathbf{P}^{\mathcal{D}}, \iota^{\mathcal{D}}_{\text{init}})$ be a DTMC. Its corresponding fixed-initial-state representation is $\mathcal{D}_{\text{uni}} = (S^{\mathcal{D}} \cup \{s_{\text{in}}\}, \mathbf{P}^{\mathcal{D}_{\text{uni}}}, s_{\text{in}})$, where

$$\mathbf{P}^{\mathcal{D}_{\text{uni}}}(s_i, s_j) = \begin{cases} \mathbf{P}^{\mathcal{D}}(s_i, s_j), & s_i, s_j \neq s_{\text{in}}, \\ \iota^{\mathcal{D}}_{\text{init}}(s_j), & s_i = s_{\text{in}}, \\ 0, & \text{otherwise}. \end{cases}$$

The two representations induce the same probability space up to the natural bijection between their paths. In particular, define

$$\phi : Paths^{\mathcal{D}_{\text{uni}}}_{\text{inf}} \longrightarrow Paths^{\mathcal{D}}_{\text{inf}}, (s_{\text{in}}, s_0, s_1, \ldots) \mapsto (s_0, s_1, \ldots).$$

ϕ is a bijection, since s_{in} has no incoming transitions, every infinite path of $\mathcal{D}_{\text{uni}}$ starts with s_{in}.

To see that the probability measures correspond under ϕ, consider a finite path $\pi_{\text{uni}} = (s_{\text{in}}, s_0, \ldots, s_n)$ of $\mathcal{D}_{\text{uni}}$, and let $\pi = (s_0, \ldots, s_n)$ be the corresponding finite path of $\mathcal{D}$. Then

$$Pr^{\mathcal{D}_{\text{uni}}}(\pi_{\text{uni}}) = \mathbf{P}^{\mathcal{D}_{\text{uni}}}(s_{\text{in}}, s_0) \prod_{k=0}^{n-1} \mathbf{P}^{\mathcal{D}_{\text{uni}}}(s_k, s_{k+1}) \quad = \iota^{\mathcal{D}}_{\text{init}}(s_0) \prod_{k=0}^{n-1} \mathbf{P}^{\mathcal{D}}(s_k, s_{k+1}) = Pr^{\mathcal{D}}(\pi).$$

Thus, corresponding cylinder sets have the same probability, and consequently the probability measures on the two path spaces coincide under the bijection ϕ.

Hence, adding a unique initial state is merely a different representation of the initial distribution and does not change the probabilistic behaviour of the model. In particular, results formulated for the fixed-initial-state definition can be transferred to our definition by applying the above construction.

To resolve the uncertainty of the initial distribution and transition probability function, we need to define semantics to interpret the Interval Markov Chain. In other words, we need to resolve an Interval Markov Chain to some set of Markov Chains that correspond to it. We say that all of these DTMCs $\mathcal{D}$ satisfy the IDTMC $\mathcal{I}$ and denote it as $\mathcal{D} \models \mathcal{I}$. We assume in this thesis that there exists at least one satisfying DTMC for every IDTMC.

Assumption 1
In this thesis, we only consider finite IDTMCs, i.e., IDTMCs that have a finite state space. Moreover, we will only consider IDTMCs where there exists at least one satisfying DTMC $\mathcal{D} \models \mathcal{I}$ for each state $s \in S^{\mathcal{I}}$, such that s is reachable with a positive probability from an initial state in $\mathcal{D}$. Because, if a given state was not reachable in any satisfying DTMC, then the EVT of this state would trivially be 0 for all satisfying DTMCs and have no impact on the EVTs of other states.

In this thesis, we will consider two popular semantics: the *once-and-for-all*, also known as the *Uncertain Markov Chain (UMC) semantic*, and the *IMDP semantic*. We will first define the former, similar to [BDLM21, CSH08, BLW13]. To this end, we fix an IDTMC $\mathcal{I} = (S^{\mathcal{I}}, \underline{\mathbf{P}}, \overline{\mathbf{P}}, \underline{\iota}_{init}, \overline{\iota}_{init})$ and a DTMC $\mathcal{D} = (S^{\mathcal{D}}, \mathbf{P}, \iota_{init})$.

Definition 4.4 (Once-and-for-all/UMC Semantic)
A DTMC $\mathcal{D}$ satisfies an IDTMC $\mathcal{I}$ if and only if $S^{\mathcal{I}} = S^{\mathcal{D}}$, and for all $s, s' \in S^{\mathcal{I}}$, $\mathbf{P}^{\mathcal{D}}(s, s') \in [\underline{\mathbf{P}}^{\mathcal{I}}(s, s'), \overline{\mathbf{P}}^{\mathcal{I}}(s, s')]$ and $\iota^{\mathcal{D}}_{init}(s) \in [\underline{\iota}^{\mathcal{I}}_{init}(s), \overline{\iota}^{\mathcal{I}}_{init}(s)]$.

In other words, any DTMC whose initial distribution and transition probabilities lie in the intervals defined by the IDTMC is satisfactory. Thus, we can interpret the *UMC* semantic for an IDTMC, as the initial distribution and probability function being chosen from the intervals *once-and-for-all*. Moreover, notice that $\mathcal{D}$ has the same cardinality as $\mathcal{I}$, i.e., $|S^{\mathcal{D}}| = |S^{\mathcal{I}}|$. If $\mathcal{D}$ satisfies $\mathcal{I}$ under the UMC semantic, we denote this as $\mathcal{D} \models_{\mathrm{U}} \mathcal{I}$. As $\mathcal{D} \models_{\mathrm{U}} \mathcal{I}$ implies $S^{\mathcal{D}} = S^{\mathcal{I}}$, we use the same symbol $s \in S^{\mathcal{D}} = S^{\mathcal{I}}$ to denote the corresponding state in both $\mathcal{D}$ and $\mathcal{I}$.

Example 4.5
Consider the IDTMC $\mathcal{I}_E$ in Fig. 4.1. The DTMC $\mathcal{D}_U = (S^{\mathcal{I}_E}, \mathbf{P}, \iota_{init})$ depicted in Fig. 4.2 satisfies $\mathcal{I}$ under the UMC semantic, i.e., $\mathcal{D}_U \models_U \mathcal{I}$, where the transition probability function is given by

$$\mathbf{P} = \begin{bmatrix} 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0.8 & 0.2 \\ 0.8 & 0 & 0 & 0 & 0.2 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix},$$

and the initial distribution

$$\iota_{init} = (1, 0, 0, 0, 0)$$

The second semantic which we will define, is the *IMDP semantic*. There are two common definitions for this semantic. In this thesis, we will use the definition found in [SVA06] and [CSH08].

While the *UMC* semantic selects a particular probability function *once-and-for-all*,

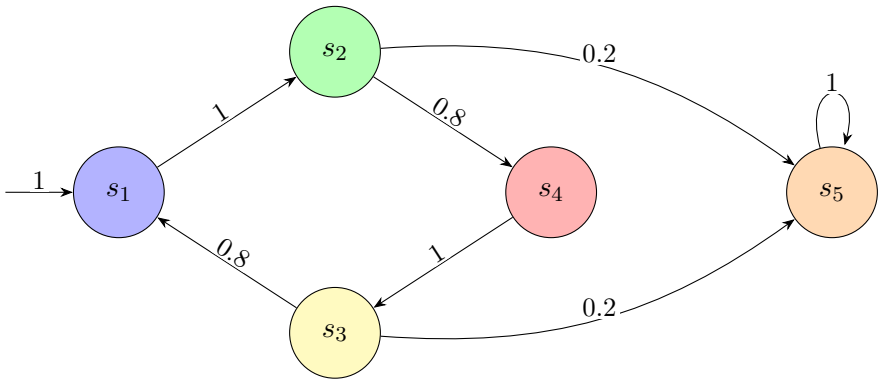


Figure 4.2: DTMC $\mathcal{D}_U$ with $\mathcal{D}_U \models_U \mathcal{I}$

the *IMDP* semantic allows for a probability function to be picked from the set of valid functions before every transition. Similar to [SVA06] and [CSH08], we define the set of valid probability functions over $S^{\mathcal{I}}$ as $Steps^{\mathcal{I}}(s)$, i.e.,

$$Steps^{\mathcal{I}}(s) = \{\mathbf{P} \mid \sum_{t \in S^{\mathcal{I}}} \mathbf{P}(s,t) = 1 \text{ and } \underline{\mathbf{P}}(s,t) \leq \mathbf{P}(s,t) \leq \overline{\mathbf{P}}(s,t) \text{ for all } t \in S^{\mathcal{I}}\}$$

for all $s \in S^{\mathcal{I}}$. Furthermore, $Steps^{\mathcal{I}} = \bigcup_{s \in S^{\mathcal{I}}} Steps^{\mathcal{I}}(s)$. In other words, under the *IMDP* semantic, a potentially different probability function can be chosen from $Steps^{\mathcal{I}}(s)$ before each time the state s is entered. This choice of a probability function is made non-deterministically by an external environment.

In [Bad] this external environment is defined as the *nature*.

Definition 4.6 (Nature)
A *nature* for an IDTMC $\mathcal{I}$ is a function $\eta : Paths^{\mathcal{I}}_{fin} \to Steps^{\mathcal{I}}$ that assigns each finite path to a valid transition probability function, i.e., $\eta(\pi) \in Steps^{\mathcal{I}}(last(\pi))$.

Notice, that we only define this external environment for the transition probability function and not the initial distribution. Recall, that results for the *fixed-initial-state definition* can be applied to our definition by applying the shown construction. Thus, it suffices to define this external environment for the transition probability function.

By applying a given *nature* η to $\mathcal{I}$, we resolve the uncertainty of the probability function, as η specifies a probability function $\mathbf{P} \in Steps^{\mathcal{I}}(s)$ for every finite path in $\mathcal{I}$. Thus, η induces a DTMC as follows. The definition is based on [BK08].

Definition 4.7 (DTMC of an IDTMC Induced by a Nature)
For an IDTMC $\mathcal{I}$ and a nature η on $\mathcal{I}$, the induced Markov Chain $\mathcal{D}_\eta$ is given by

$$\mathcal{D}_\eta = (Paths^{\mathcal{I}}_{fin}, \mathbf{P}^{\mathcal{D}_\eta}, \iota^{\mathcal{I}}_{init}) \ ,$$

where for all $\pi \in Paths^{\mathcal{I}}_{fin}$, $s \in S^{\mathcal{I}}$, and $\eta(\pi) = \mathbf{P}(last(\pi))$:

$$\mathbf{P}^{\mathcal{D}_\eta}(\pi, \pi') = \begin{cases} \mathbf{P}(last(\pi), s) & \pi' = (\pi[0], \pi[1], \dots, last(\pi), s) \\ 0 & \text{otherwise.} \end{cases}$$

Notice, that these induced DTMCs are infinite in size, i.e., $|\mathcal{D}_\eta| = \infty$ and, that intuitively, every state $s \in S^{\mathcal{D}_\eta}$ is visited only once, as the states represent paths. Thus, $\mathcal{D}_\eta$ is an acyclic DTMC. To relate the states in the induced DTMCs $\mathcal{D}_\eta$ to the states in the IDTMC $\mathcal{I}$, we define the function

$$\tau : Paths^{\mathcal{I}}_{fin} \to S^{\mathcal{I}}, \ (\pi[0], \dots, last(\pi)) \mapsto last(\pi) \ .$$

In other words, the states in the induced DTMCs that correspond to paths ending in some state $s \in S^{\mathcal{I}}$, are mapped to s. Intuitively,

$$\tau^{-1}(s) = \{\pi \in Paths^{\mathcal{I}}_{fin} \mid \tau(\pi) = s\}$$

defines the set of all states in the induced DTMCs that correspond to paths ending in s for all $s \in S^{\mathcal{I}}$. Moreover, we denote $\tau^{-1}(T) = \bigcup_{t \in T} \tau^{-1}(t)$ for $T \subseteq S^{\mathcal{I}}$.

Similarly, we define the set

$$\rho(s) = \{\pi \in Paths^{\mathcal{I}}_{fin} \mid last(\pi) = s \wedge \nexists k \in \underline{|\pi| - 1} : \pi[k] = s\} \ .$$

In other words, this set contains all states in $\mathcal{D}_\eta$ that correspond to finite paths in $\mathcal{I}$ which end in s and contain s precisely once. We denote $\rho(T) = \bigcup_{t \in T} \rho(t)$ for $T \subseteq S^{\mathcal{I}}$.

Importantly, however, there exist certain types of natures, see Def. 4.8, that can be represented by finite-state DTMCs.

One type of such nature, which is particularly relevant for this thesis, is the *stationary nature*. The following definition is based on [Bad].

Definition 4.8 (Stationary Nature)
A nature η for an IDTMC $\mathcal{I}$ is *stationary* if for all paths $\pi, \pi' \in Paths^{\mathcal{I}}_{fin}$, it holds that

$$last(\pi) = last(\pi') \implies \eta(\pi) = \eta(\pi') \ .$$

Intuitively, this implies that η only depends on the current state of the model, i.e., the state the model is in. Thus, stationary natures are functions of the form

$$\eta : S^{\mathcal{I}} \to Steps^{\mathcal{I}} \ .$$

Notice that the induced DTMCs of these stationary natures are finite in size.

Definition 4.9 (DTMC of an IDTMC Induced by a Stationary Nature)
For an IDTMC $\mathcal{I}$ and a stationary nature η on $\mathcal{I}$, the induced Markov Chain $\mathcal{D}_\eta$ is given by

$$\mathcal{D}_\eta = (S^{\mathcal{I}}, \mathbf{P}^{\mathcal{D}_\eta}, \iota^{\mathcal{I}}_{init}) \ ,$$

where for all $s_i, s_j \in S^{\mathcal{I}}$ and $\eta(s_i) = \mathbf{P}(s_i)$:

$$\mathbf{P}^{\mathcal{D}_\eta}(s_i, s_j) = \mathbf{P}(s_i, s_j)$$

Crucially, a stationary nature can thus be represented by a DTMC of the form Def. 4.9 as well as Def. 4.7. However, as all natures can be represented by the DTMC provided in Def. 4.7, we will base all defintions for IDTMCs under the IMDP semantic in this thesis on Def. 4.7.

Using these notions, we can now define the *IMDP semantic*.

Definition 4.10 (IMDP semantic)
A DTMC $\mathcal{D}$ satisfies an IDTMC $\mathcal{I}$ if and only if it is induced by a *nature* $\eta : Paths^{\mathcal{I}}_{fin} \to Steps^{\mathcal{I}}$ on $\mathcal{I}$.

If $\mathcal{D}$ satisfies $\mathcal{I}$ under the IMDP semantic, we denote this as $\mathcal{D} \models_{\mathrm{I}} \mathcal{I}$.

The other common definition can be found in [BDL+17] and [BDLM21].

Example 4.11
Consider the IDTMC $\mathcal{I}$ in Fig. 4.1. The DTMC $\mathcal{D}_I$ depicted in Fig. 4.3 satisfies $\mathcal{I}$ under the IMDP semantic, i.e., $\mathcal{D}_I \models_I \mathcal{I}$, where the transition probability function is given by

$$\mathbf{P} = \begin{bmatrix} 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0.8 & 0.2 \\ 0.8 & 0 & 0 & 0 & 0.2 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix} ,$$

and the initial distribution

$$\iota_{init} = (0.6, 0, 0, 0, 0)$$

If we want to emphasize that a DTMC $\mathcal{D}$ satisfies the IDTMC $\mathcal{I}$ under both semantics, we write $\mathcal{D} \models \mathcal{I}$. Furthermore, if two DTMCs $\mathcal{D}_1$ and $\mathcal{D}_2$ satisfy a given IDTMC $\mathcal{I}$, we write $\mathcal{D}_1, \mathcal{D}_2 \models \mathcal{I}$ for short.

Lastly, we introduce some notions of IDTMCs, which we use in the following chapters.

First, we introduce reachability probabilities for IDTMCs. We base our definition of the quantitative reachability property in IDTMCs on [BDL+17].

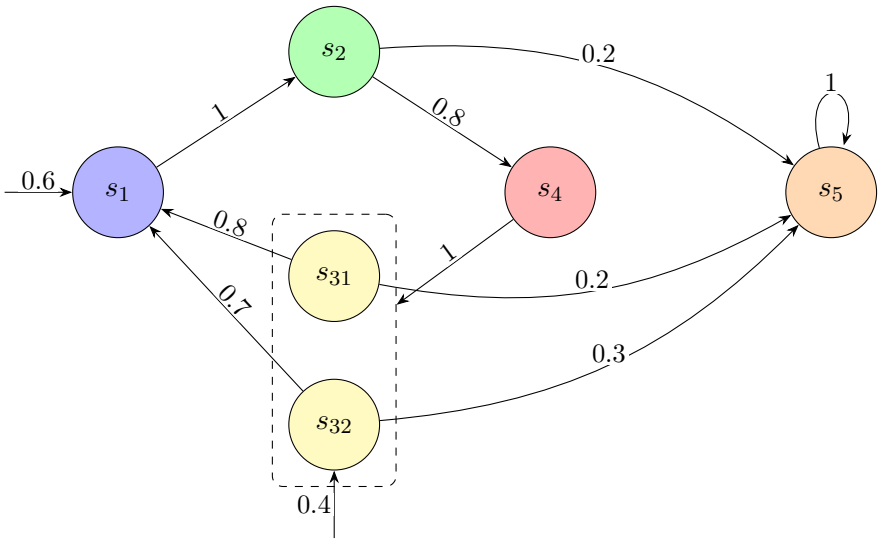


Figure 4.3: DTMC $\mathcal{D}_I$ with $\mathcal{D}_I \models_I \mathcal{I}$.

Definition 4.12 (Reachability Probability in IDTMCs)
Given an IDTMC $\mathcal{I}$, a set of states $T \subseteq S$, and $p \in [0,1]$, then a quantitative reachability property on $\mathcal{I}$ under the UMC semantic is defined as $Pr^{\mathcal{I}}(\Diamond T) = p$, if and only if there exists at least one DTMC $\mathcal{D} \models_U \mathcal{I}$ such that $Pr^{\mathcal{D}}(\Diamond T) = p$. Further, we define the quantitative reachability property on $\mathcal{I}$ under the IMDP semantic as $Pr^{\mathcal{I}}(\Diamond T) = p$, if and only if there exists at least one DTMC $\mathcal{D} \models_I \mathcal{I}$ such that $Pr^{\mathcal{D}}(\bigcup_{t \in \rho(T)} \Diamond t) = p$ under the IMDP semantic.

Additionally, we use the notation $Pr_s^{\mathcal{I}}(\Diamond T)$ to denote the reachability probability, when fixing $s \in S^{\mathcal{I}}$ as the unique initial state.

In this thesis, we are particularly interested in the optimal reachability probabilities, which we define as follows.

Definition 4.13 (Maxmimum and Minimum Reachability Probability)
Given an IDTMC $\mathcal{I}$ and a set of states $T \subseteq S$, we define the maximum reachability as $\sup_{\mathcal{D} \models_U \mathcal{I}} Pr^{\mathcal{D}}(\Diamond T)$ and the minimum reachability as $\inf_{\mathcal{D} \models_U \mathcal{I}} Pr^{\mathcal{D}}(\Diamond T)$ und the UMC semantic.

Furthermore, we define the maximum reachability as $\sup_{\mathcal{D} \models_I \mathcal{I}} Pr^{\mathcal{D}}(\bigcup_{t \in \rho(T)} \Diamond t)$ and the minimum reachability as $\inf_{\mathcal{D} \models_I \mathcal{I}} Pr^{\mathcal{D}}(\bigcup_{t \in \rho(T)} \Diamond t)$ und the IMDP semantic.

To shorten the notation we will denote the maximum reachability as $\overline{Pr}^{\mathcal{D}}(\Diamond s)$ and the minimum reachability as $\underline{Pr}^{\mathcal{D}}(\Diamond s) = \inf_{\mathcal{D}_i \models \mathcal{I}} Pr^{\mathcal{D}_i}(\Diamond s)$. Moreover, we use the notation $\overline{Pr}_s^{\mathcal{I}}(\Diamond T)$ and $\underline{Pr}_s^{\mathcal{I}}(\Diamond T)$ to denote the maximum and minimum reachability

probabilities, when fixing $s \in S^{\mathcal{I}}$ as the unique initial state.

Furthermore, we define transient and recurrent states. As we observed for the infinite-state DTMCs $\mathcal{D}_\eta$ defined in Def. 4.7, every state is visited only once. Thus, intuitively $Pr_s(\bigcirc\Diamond s) = 0$ for all $s \in S^{\mathcal{D}_\eta}$, meaning $S^{\mathcal{D}_\eta}$ contains no recurrent states. Consequently, we define transient and recurrent states exclusively for the UMC semantic.

Definition 4.14 (Transient and Recurrent states in IDTMCs)
Given an IDTMC $\mathcal{I}$, a state $s \in S^{\mathcal{I}}$ is called recurrent if and only if $\overline{Pr}_s^{\mathcal{I}}(\bigcirc\Diamond s) = 1$, i.e., there exists at least one satisfying DTMC $\mathcal{D} \models_U \mathcal{I}$ such that $Pr_s^{\mathcal{D}}(\bigcirc\Diamond s) = 1$.

Furthermore, a state $s \in S^{\mathcal{I}}$ is called transient if and only if $\underline{Pr}_s^{\mathcal{I}}(\bigcirc\Diamond s) < 1$, i.e., there exists at least one satisfying DTMC $\mathcal{D} \models_U \mathcal{I}$ such that $Pr_s^{\mathcal{D}}(\bigcirc\Diamond s) < 1$.

We denote the set of recurrent states as $S_{re}^{\mathcal{I}}$ and the set of transient states as $S_{tr}^{\mathcal{I}}$.

Notice, that in contrast to DTMCs, there exist IDTMCs $S_{tr}^{\mathcal{I}} \cap S_{re}^{\mathcal{I}} \neq \emptyset$. To observe this, consider state s_4 of Example 4.2. Let $\mathcal{D}_{tr} \models_U \mathcal{I}_E$ with $\mathbf{P}^{\mathcal{D}_{tr}}(s_4, s_3) = 1$. Therefore, $s_4 \in S_{tr}^{\mathcal{D}_{tr}}$ and consequently $s_4 \in S_{tr}^{\mathcal{I}_E}$. Now, let $\mathcal{D}_{re} \models_U \mathcal{I}_E$ with $\mathbf{P}^{\mathcal{D}_{re}}(s_4, s_4) = 1$. Then, $s_4 \in S_{re}^{D_{re}}$ and thus $s_4 \in S_{re}^{\mathcal{I}_E}$. Thus, $s_4 \in S_{re}^{\mathcal{I}_E} \cap S_{tr}^{\mathcal{I}_E}$.

Due to this, we additionally define the sets $S_{\forall re} = \{s \in S^{\mathcal{I}} \mid \forall \mathcal{D} \models_U \mathcal{I} : Pr_s^{\mathcal{D}}(\bigcirc\Diamond s) = 1 \wedge Pr^{\mathcal{D}}(\Diamond s) > 0\}$ and $S_{\forall tr} = \{s \in S^{\mathcal{I}} \mid \forall \mathcal{D} \models_U \mathcal{I} : Pr_s^{\mathcal{D}}(\bigcirc\Diamond s) < 1 \wedge Pr^{\mathcal{D}}(\Diamond s) > 0\}$. Intuitively, these are the sets that contain all states that are either recurrent or transient, respectively, and reachable in all DTMCs $\mathcal{D} \models_U \mathcal{I}$ that satisfy $\mathcal{I}$ under the UMC semantic.

4.1 Relation between the semantics

Intuitively, the *IMDP* semantic is more expressive than the *UMC* semantic. More specifically for any given IDTMC $\mathcal{I}$, a DTMC $\mathcal{D}$ that satisfies $\mathcal{I}$ according to the UMC semantic, i.e., $\mathcal{D} \models_U \mathcal{I}$, does also satisfy $\mathcal{I}$ according to the IMDP semantic, i.e., $\mathcal{D} \models_I \mathcal{I}$.

Theorem 4.15
For any IDTMC $\mathcal{I}$, if $\mathcal{D} \models_U \mathcal{I}$, then $\mathcal{D} \models_I \mathcal{I}$.

We can prove this theorem using the following lemma.

Lemma 4.16
Given an IDTMC $\mathcal{I}$ and stationary nature η, for every induced DTMC $\mathcal{D}_\eta$, there exists a DTMC $\mathcal{D} \models_U \mathcal{I}$, such that $\mathcal{D} = \mathcal{D}_\eta$, i.e., $\mathcal{D}_\eta \models_U \mathcal{I}$.

In other words the set of induced DTMCs by stationary natures on $\mathcal{I}$ is identical to

the set of DTMCs that satisfy $\mathcal{I}$ under the UMC semantic.

Proof 7
First notice, that $S^{\mathcal{D}_\eta} = S^{\mathcal{I}}$ by definition, see Def. 4.9. Moreover, $\mathbf{P}^{\mathcal{D}_\eta}(s_i, s_j) = \mathbf{P}(s_i, s_j)$ for all $s_i, s_j \in S^{\mathcal{I}}$, where $\mathbf{P} = \eta(s_i)$. As aforementioned, a stationary nature is a function $\eta : S^{\mathcal{I}} \to Steps^{\mathcal{I}}$. Thus, $\mathbf{P}(s_i) \in Steps(s_i)^{\mathcal{I}}$ and therefore $\sum_{t \in S^{\mathcal{I}}} \mathbf{P}(s_i, t) = 1$, i.e., $\mathbf{P}$ is well-defined, and $\mathbf{P}(s_i, t) \in [\underline{\mathbf{P}}(s_i, t), \overline{\mathbf{P}}(s_i, t)]$ for all $t \in S^{\mathcal{I}}$. It suffices to show that the condition for the transition probability function is satisfied, as we can apply results for the *fixed-intial-state-definition* to our own definition. □

As stationary natures are a specific type of nature, Theorem 4.15 follows directly from Lemma 4.16.

Another important relation between the two semantics, that we use in this thesis, is that the semantics coincide on the maximum and minimum reachability probability, see [BDL+17, CHK13].

Chapter 5

EVTs in IDTMCs

We continue by defining *expected visiting times* (EVTs) for IDTMCs. As we discussed in the previous section, IDTMCs can be seen as a set of DTMCs that satisfy the IDTMC according to the given semantic. These different DTMCs can have varying EVTs for a given state s in the IDTMC, thus we are interested in the upper and lower bound for the EVTs of s, i.e., in the highest and lowest EVT of s among the satisfying DTMCs.

Definition 5.1 (EVTs of IDTMCs)
For a IDTMC $\mathcal{I}$, we define the *upper expected visiting time* (uEVT) $\overline{\mathbb{E}}^{\mathcal{I}}[vt_s]$ and *lower expected visiting time* (ℓEVT) $\underline{\mathbb{E}}^{\mathcal{I}}[vt_s]$ for all states $s \in S^{\mathcal{I}}$, as follows for the UMC semantic

(i) $\underline{\mathbb{E}}^{\mathcal{I}}[vt_s] = \inf_{\mathcal{D} \models_U \mathcal{I}} \mathbb{E}^{\mathcal{D}}[vt_s]$

(ii) $\overline{\mathbb{E}}^{\mathcal{I}}[vt_s] = \sup_{\mathcal{D} \models_U \mathcal{I}} \mathbb{E}^{\mathcal{D}}[vt_s]$

and the IMDP semantic

(i) $\underline{\mathbb{E}}^{\mathcal{I}}[vt_s] = \inf_{\mathcal{D} \models_I \mathcal{I}} \mathbb{E}^{\mathcal{D}}[vt_T]$

(ii) $\overline{\mathbb{E}}^{\mathcal{I}}[vt_s] = \sup_{\mathcal{D} \models_I \mathcal{I}} \mathbb{E}^{\mathcal{D}}[vt_T]$,

where $T = \tau^{-1}(s)$.

5.1 Calculating Expected Visiting Times of IDTMCs

We will now have a look at how the *expected visiting times* for IDTMCs can be calculated.

The most intuitive upper and lower bounds for the uEVTs and lEVTs of IDTMCs can be derived by using Eq. (3.2) and the upper and lower transition probability functions

and initial distributions defined in Def. 4.1.

Theorem 5.2 (Upper and Lower Bounds for uEVTs and lEVTs)
For any IDTMC $\mathcal{I}$, the uEVT is bounded from above by

$$\overline{\mathbb{E}}[vt_s] \leq \mathbb{E}_{upp}[vt_s] \coloneqq \overline{\iota}_{init}(s) + \sum_{t \in S_{tr}} \overline{\mathbf{P}}(t,s) \cdot \mathbb{E}_{upp}[vt_t]$$

and the ℓEVT is bounded from below by

$$\underline{\mathbb{E}}[vt_s] \geq \mathbb{E}_{low}[vt_s] = \underline{\iota}_{init}(s) + \sum_{t \in S_{tr}} \underline{\mathbf{P}}(t,s) \cdot \mathbb{E}_{low}[vt_t]$$

for all $s \in S^{\mathcal{I}}$.

Proof 8
For any IDTMC $\mathcal{I}$ there exists a DTMC $\mathcal{D}$ that satisfies $\mathcal{I}$, i.e., $\mathcal{D} \models \mathcal{I}$, and $\mathbb{E}^{\mathcal{D}}[vt_s] = \overline{\mathbb{E}}^{\mathcal{I}}[vt_s]$. As $\mathcal{D} \models \mathcal{I}$, it follows that $\mathbf{P}(s,s') \leq \overline{\mathbf{P}}(s,s')$ for all $s, s' \in S$.

Thus, $\mathbb{E}^{\mathcal{D}}[vt_s] = \iota_{init}(s) + \sum_{t \in S_{tr}} \mathbf{P}(t,s)^{\mathsf{T}} \cdot \mathbb{E}[vt_t] \leq \overline{\iota}_{init}(s) + \sum_{t \in S_{tr}} \overline{\mathbf{P}}(t,s)^{\mathsf{T}} \cdot \mathbb{E}_{upp}[vt_t] = \mathbb{E}^{\mathcal{I}}_{upp}[vt_s]$.

The identical argument can be made to prove the lower bound for the ℓEVT. □

As we fix the transition probability functions to either $\underline{\mathbf{P}}$ or $\overline{\mathbf{P}}$, we remove the uncertainty of the IDTMCs and obtain a fixed transition probability function. Notice, that intuitively $\underline{\mathbf{P}}$ is row-substochastic. Thus, we can calculate $\mathbb{E}_{low}$ by using Eq. (3.2) in PTIME, as argued in Sec. 3.2. The same, however, does not hold for $\overline{\mathbf{P}}$

Moreover, these bounds are not tight, as can be seen in the following example.

Example 5.3
For this take a look at the upper bound of the IDTMC provided in Fig. 5.1. Using Theorem 5.2, we can calculate that $\mathbb{E}_{upp}[vt_{s_4}] = 2$, as $\mathbb{E}_{upp}[vt_{s_1}] = \mathbb{E}_{upp}[vt_{s_2}] = \mathbb{E}_{upp}[vt_{s_3}] = 1$. However, as the IDTMC is acyclic, see Def. 3.10, every state can be visited at most once. Thus, the upper bound $\mathbb{E}_{upp}[vt_{s_4}]$ is clearly not tight. Similarly, $\mathbb{E}_{low}[vt_{s_4}] = 0$, however s_4 needs to be visited at least once. Thus, this lower bound $\mathbb{E}_{low}[vt_{s_4}]$ is not tight either.

In the following, we will therefore take a look at how we can characterize the precise values of the uEVTs and ℓEVTs.

First, we provide some characterizations under the UMC semantic, as our definition of transient and recurrent states in IDTMCs is limited to the UMC semantic, see Def. 4.14.

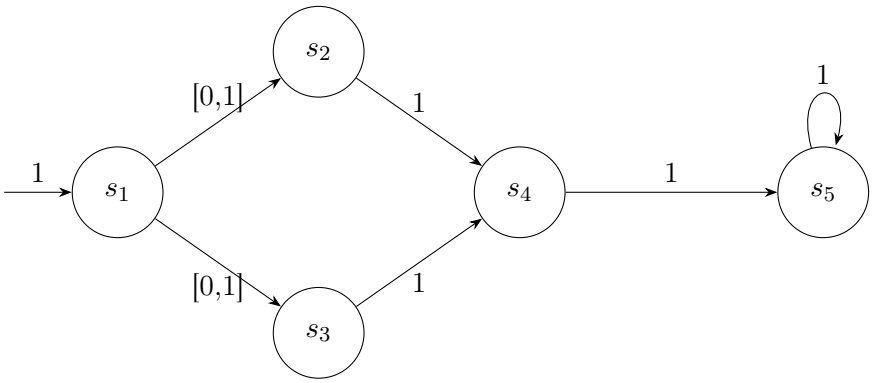


Figure 5.1: An example of a DTMC.

Theorem 5.4
For a given state s in an IDTMC $\mathcal{I}$, $\overline{\mathbb{E}}[vt_s] = \infty$ if and only if $s \in S^{\mathcal{I}}_{re}$. Moreover, $\overline{\mathbb{E}}[vt_s] = \underline{\mathbb{E}}[vt_s] = \infty$ if and only if $s \in S^{\mathcal{I}}_{\forall re}$.

Proof 9
By our Assumption 1, there exists a DTMC $\mathcal{D} \models_U \mathcal{I}$ such that s is reachable in $\mathcal{D}$, i.e., $Pr^{\mathcal{D}}(\Diamond s) > 0$ and $\overline{Pr}^{\mathcal{I}}(\Diamond s) > 0$. Moreover, as $s \in S^{\mathcal{I}}_{re}$, there exists a DTMC $\mathcal{D}_{re} \models_U \mathcal{I}$ such that $Pr^{\mathcal{D}_{re}}_s(\bigcirc\Diamond s) = 1$, see Def. 4.14, thus $\overline{Pr}^{\mathcal{I}}_s(\bigcirc\Diamond s) = 1$. Using Theorem 5.11, there exists a DTMC $\mathcal{D}_{+re} \models \mathcal{I}$ such that $Pr^{\mathcal{D}_{+re}}(\Diamond s) > 0$ and $Pr^{\mathcal{D}_{+re}}_{N(s)}(\Diamond s) = 1$, as $\overline{Pr}^{\mathcal{I}}(\Diamond s) > 0$ and $\overline{Pr}^{\mathcal{I}}_s(\bigcirc\Diamond s) = 1$. Therefore, clearly $s \in S^{\mathcal{D}_{+re}}_{re}$, see Def. 3.11 and using Lemma 3.18, we can thus conclude that $\mathbb{E}^{\mathcal{D}_{+re}}[vt_s] = \infty$ and thus $\overline{\mathbb{E}}^{\mathcal{I}}[vt_s] = \infty$.

Conversely, if $\overline{\mathbb{E}}^{\mathcal{I}}[vt_s] = \infty$, then there exists a DTMC $\mathcal{D} \models \mathcal{I}$ such that $\mathbb{E}^{\mathcal{D}}[vt_s] = \infty$, i.e. $s \in S^{\mathcal{D}}_{re}$, see Lemma 3.18. Thus, it follows directly that $s \in S^{\mathcal{I}}_{re}$, see Def. 4.14.

Moreover, $\overline{\mathbb{E}}[vt_s] = \underline{\mathbb{E}}[vt_s] = \infty$ if and only if $s \in S^{\mathcal{I}}_{\forall re}$ as $Pr^{\mathcal{D}}(\Diamond s) > 0$ and $Pr^{\mathcal{D}}_s(\bigcirc\Diamond s) > 0$ for all $\mathcal{D} \models_U \mathcal{I}$, by the definition of $S_{\forall re}$. Thus, using Lemma 3.18 we can directly conclude the proof. □

Theorem 5.5
For any $s \in S^{\mathcal{I}}$ in a given IDTMC $\mathcal{I}$, $\underline{\mathbb{E}}[vt_s] = 0$ if and only if $\underline{Pr}^{\mathcal{I}}(\Diamond s) = 0$, i.e., there exists a satisfying DTMC of $\mathcal{I}$, such that s is not reachable.

Proof 10
If $\underline{Pr}^{\mathcal{I}}(\Diamond s) = 0$, then there exists $\mathcal{D} \models \mathcal{I}$ such that $Pr^{\mathcal{D}}(\Diamond s) = 0$, i.e., there exists no path in $\mathcal{D}$ that contains s. As therefore, $vt_s(\pi) = 0$ for all $\pi \in Paths^{\mathcal{D}}_{inf}$, we can conclude, using Def. 3.17, that $\mathbb{E}^{\mathcal{D}}[vt_s] = 0$ and thus $\underline{\mathbb{E}}^{\mathcal{I}}[vt_s] = 0$.

Conversely, if $\underline{Pr}^{\mathcal{I}}(\Diamond s) > 0$, then there exists $\pi \in Paths^{\mathcal{D}}_{inf}$ with $vt_s(\pi) > 0$ for all $\mathcal{D} \models \mathcal{I}$. Therefore, $\mathbb{E}^{\mathcal{D}}[vt_s] > 0$ and thus $\underline{\mathbb{E}}^{\mathcal{I}}[vt_s] > 0$. □

Remark. *Non-attainability of values in EVT intervals under the UMC semantic.*
In general, not every value in $[\underline{\mathbb{E}}[vt_s], \overline{\mathbb{E}}[vt_s]]$ is attainable under the UMC semantic. In other words, there exists an IDTMC with state s, such that there exists a $x \in [\underline{\mathbb{E}}^{\mathcal{I}}[vt_s], \overline{\mathbb{E}}^{\mathcal{I}}[vt_s]]$ with $\mathbb{E}^{\mathcal{D}}[vt_s] \neq x$ for all $\mathcal{D} \models_U \mathcal{I}$. This holds even for $\overline{\mathbb{E}}[vt_s] < \infty$.

We provide a short example for this remark.

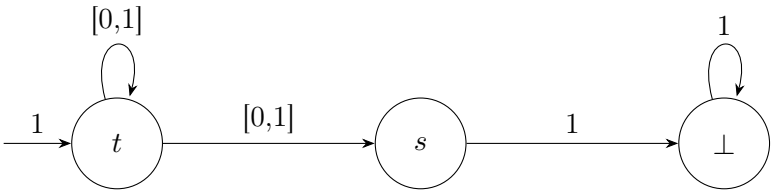


Figure 5.2: An example of a DTMC.

Example 5.6
Consider the IDTMC $\mathcal{I} = (\{t, s, \bot\}, \underline{\mathbf{P}}, \overline{\mathbf{P}}, \underline{\iota}_{init}, \overline{\iota}_{init})$ shown in Fig. 5.2. Clearly, $\underline{\mathbb{E}}^{\mathcal{I}}[vt_s] = 0$ and $\overline{\mathbb{E}}^{\mathcal{I}}[vt_s] = 1$. However, there exists no $\mathcal{D} \models_U \mathcal{I}$ such that $\mathbb{E}^{\mathcal{D}}[vt_s] = x$ for any $x \in (0, 1)$, as $\mathbb{E}^{\mathcal{D}}[vt_s] = 1$ if $\mathbf{P}^{\mathcal{D}}(t, s) > 0$ and $\mathbb{E}^{\mathcal{D}}[vt_s] = 0$ else.

5.2 Relating Expected Visiting Times and Quantitative Reachability

Similar to Sec. 3.3, we now relate *expected visiting times* to *quantitative reachability*. Moreover, we show that for any IDTMC $\mathcal{I}$, there exists a DTMC $\mathcal{D} \models \mathcal{I}$ that has the optimal, i.e., maximum or minimum, reachability probability for all states in $\mathcal{I}$.

Given the quantitative reachabilities of the DTMCs that satisfy a certain IDTMC $\mathcal{I}$, we can calculate the corresponding EVTs of a given state s in $\mathcal{I}$ and thus find the lEVTs and uEVTs using the results from Sec. 3.3.

As we concluded in Sec. 3.3, we can calculate the EVT for all transient states $s \in S^{\mathcal{D}}_{tr}$ in a DTMC $\mathcal{D}$ via quantitative reachability, using Eq. (3.6): $\mathbb{E}^{\mathcal{D}}[vt_s] = \sum_{t \in S_{tr} \cap in} \frac{Pr^{\mathcal{D}}_t(\Diamond s)}{1 - Pr^{\mathcal{D}}_s(\bigcirc \Diamond s)} \cdot \iota_{init}(t)$. As defined in Def. 5.1, the uEVT and ℓEVT, under the UMC semantic, are

equal to the maximum and minimum EVT. Thus, the uEVT and ℓEVT are given by the largest and smallest ratio of these quantitative reachability probabilites among the satisfying Markov Chains, as those optimal ratios define the maximum and minimum EVT.

In the rest of this section, we prove that for all states $s \in S^{\mathcal{I}}$ there exists at least one DTMC $\mathcal{D} \models_U \mathcal{I}$ such that $\mathcal{D}$ has maximal reachability probability for all states $t \in S^{\mathcal{I}}$ to s, i.e., $Pr_t^{\mathcal{D}}(s) = \overline{Pr}_t^{\mathcal{I}}(s)$. Conversely, we show the same for the minimum reachability probability.

Remark. *Existence of DTMC with optimal reachability.* For any finite IDTMC $\mathcal{I}$, there exists an optimal stationary nature η on $\mathcal{I}$ which induces a DTMC $\mathcal{D}_\eta$ such that $Pr_t^{\mathcal{D}_\eta}(\Diamond s) = \overline{Pr}_t^{\mathcal{I}}(\Diamond s)$ for all $t, s \in S^{\mathcal{I}}$, see [BK08, Lemma 10.102] and [Bad]. As $\mathcal{D}_\eta \models \mathcal{I}$, i.e., $\mathcal{D}_\eta$ satisfies $\mathcal{I}$ under both the UMC and IMDP semantic, see Lemma 4.16, there thus exists a DTMC $\mathcal{D} \models \mathcal{I}$ with $Pr_t^{\mathcal{D}}(\Diamond s) = \overline{Pr}_t^{\mathcal{I}}(\Diamond s)$ for every $t, s \in S^{\mathcal{I}}$ which is precisely $\mathcal{D} = \mathcal{D}_\eta$. The same holds for the minimum reachability probability, see [BK08, Lemma 10.113].

Intermediate States. Using the notion of finite paths, we introduce *intermediate states*.

Definition 5.7 (Intermediate States)
Given a DTMC $\mathcal{D}$, a set of states T, and a state $s \in S$, then $u \in S$ is *intermediate* if and only if there exists path $\pi \in Paths_{fin}^{\mathcal{D}}$ with $|\pi| = n$, such that $\pi[0] \in T$, $\pi[n] = s$, and $\pi[k] = u$ for some $k \in \underline{n}$.

In other words, there exists a finite path from some $t \in T$ to s in $\mathcal{D}$ that visits u. We denote the set of these intermediate states as $I_{T,s}^{\mathcal{D}}$ for a set of states $T \in S$ and state $s \in S$. Moreover, if T is a singleton set, i.e., $T = \{t\}$, then we simply denote the set of intermediate states as $I_{t,s}^{\mathcal{D}}$.

Theorem 5.8
Given an IDTMC $\mathcal{I}$, states $t, s \in S^{\mathcal{I}}$, and a DTMC $\mathcal{D}$ with $\mathcal{D} \models_U \mathcal{I}$.

If $\mathcal{D}$ has the maximum reachability probability from t to s, i.e., $Pr_t^{\mathcal{D}}(\Diamond s) = \overline{Pr}_t^{\mathcal{I}}(\Diamond s)$, then for all $u \in I_{t,s}^{\mathcal{D}}$:

$$Pr_u^{\mathcal{D}}(\Diamond s) = \overline{Pr}_u^{\mathcal{I}}(\Diamond s)$$

Conversely, if $\mathcal{D}$ has the minimum reachability probability from t to s, i.e., $Pr_t^{\mathcal{D}}(\Diamond s) = \underline{Pr}_t^{\mathcal{I}}(\Diamond s)$, then for all $u \in I_{t,s}^{\mathcal{D}}$:

$$Pr_u^{\mathcal{D}}(\Diamond s) = \underline{Pr}_u^{\mathcal{I}}(\Diamond s)$$

In other words, if the reachability probability, under the UMC semantic, from t to s is optimal, then the reachability probability of all intermediate states is optimal as well.

Proof 11
As stated in the previous remark, there exists at least one DTMC $\mathcal{D}_\eta \models_U \mathcal{I}$ with $Pr_t^{\mathcal{D}}(\Diamond s) = \overline{Pr}_t^{\mathcal{I}}(\Diamond s)$ for every $t, s \in S^{\mathcal{I}}$, such that $\mathcal{D}_\eta$ is induced by a stationary nature η on $\mathcal{I}$.

Based on [BK08, Theorem 10.100], we can formulate the following equation which characterizes the maximum reachability probability of IDTMCs under the IMDP semantic.

$$\overline{Pr}_t^{\mathcal{I}}(\Diamond s) = \max_{\mathbf{P} \in Steps^{\mathcal{I}}(t)} \left\{ \sum_{u \in N(t)} \mathbf{P}(t,u) \cdot \overline{Pr}_u^{\mathcal{I}}(\Diamond s) \right\} . \tag{5.1}$$

Intuitively, the DTMCs $\mathcal{D}_{\eta'}$ that are induced by natures η' which solve this equation have the maximum reachability probability $Pr_u^{\mathcal{D}_{eta}}(\Diamond s) = \overline{Pr}_t^{\mathcal{I}}(\Diamond s)$ for all $u \in I_{t,s}^{\mathcal{D}}$, as shown in [BK08]. As $\mathcal{D}_\eta$ has the maximum reachability probability of t, when starting in s, and $\mathcal{D}_\eta \models_I$, see theorem 4.15, $\mathcal{D}_\eta$ solves Eq. (5.1).

Conversely, the same can be reasoned for minimum reachability using [BK08, Theorem 10.109]. □

Corollary 5.9
Given an IDTMC $\mathcal{I}$, two transient states $t_1, t_2 \in S^{\mathcal{I}}$, a (goal) state $s \in S^{\mathcal{I}}$, and two DTMCs $\mathcal{D}_1, \mathcal{D}_2$ with $\mathcal{D}_1, \mathcal{D}_2 \models_U \mathcal{I}$ and $Pr_{t_1}^{\mathcal{D}_1}(\Diamond s) = \overline{Pr}_{t_1}^{\mathcal{I}}(\Diamond s)$ and $Pr_{t_2}^{\mathcal{D}_2}(\Diamond s) = \overline{Pr}_{t_2}^{\mathcal{I}}(\Diamond s)$. In other words the DTMCs have a maximum reachability probability for t_1 and t_2, respectively.

For all $u \in I_{t_1,s}^{\mathcal{D}_1} \cap I_{t_2,s}^{\mathcal{D}_2}$, the reachability probabilities $Pr_u^{\mathcal{D}_1}(\Diamond s) = Pr_u^{\mathcal{D}_2}(\Diamond s)$ are identical.

Conversely, the same is true for minimum reachability probabilities.

In other words, if $\mathcal{D}_1$ and $\mathcal{D}_2$ have optimal reachability from t_1 and t_2, respectively, to s, then all states $u \in S^{\mathcal{I}}$ that are intermediate states of t_1 and s as well as t_2 and s have the identical reachability probability in $\mathcal{D}_1$ and $\mathcal{D}_2$.

Proof 12
This corollary follows from Theorem 5.8. Using Theorem 5.8, we can conclude that $Pr_u^{\mathcal{D}_1}(\Diamond s) = \overline{Pr}_u^{\mathcal{I}}(\Diamond s)$ for all $u \in I_{t_1,s}^{\mathcal{D}_1}$ and $Pr_u^{\mathcal{D}_2}(\Diamond s) = \overline{Pr}_u^{\mathcal{I}}(\Diamond s)$ for all $u \in I_{t_2,s}^{\mathcal{D}_2}$. Thus, $Pr_u^{\mathcal{D}_1}(\Diamond s) = Pr_u^{\mathcal{D}_2}(\Diamond s)$ for all $u \in I_{t_1,s}^{\mathcal{D}_1} \cap I_{t_2,s}^{\mathcal{D}_2}$.

Conversely, this holds for minimum reachability probabilities using the same reasoning. □

Lemma 5.10 (One-state Extension)
Given a set $T \subset S$, a state $t \notin T$, and a state s, assume that $\mathcal{D}_T \models_U \mathcal{I}$ with $Pr_u^{\mathcal{D}_T}(\Diamond s) = \overline{Pr}_u^{\mathcal{I}}(\Diamond s)$ for all $u \in T$ and $\mathcal{D}_t \models_U \mathcal{I}$ with $Pr_t^{\mathcal{D}_t}(\Diamond s) = \overline{Pr}_t^{\mathcal{I}}(\Diamond s)$. Define $\mathcal{D}_{cmb} = (S^{\mathcal{I}}, \mathbf{P}^{\mathcal{D}_{cmb}}, \iota_{init}^{\mathcal{D}_T})$ by

$$\mathbf{P}^{\mathcal{D}_{cmb}}(w,v) = \begin{cases} \mathbf{P}^{\mathcal{D}_t}(w,v) & \text{if } w \in I_{t,s}^{\mathcal{D}_t} \setminus I_{T,s}^{\mathcal{D}_T} \\ \mathbf{P}^{\mathcal{D}_T}(w,v) & \text{else.} \end{cases}$$

Stated in words, $\mathcal{D}_{cmb}$ is constructed such that its probability distribution function is equal to that of $\mathcal{D}_t$ in all states that are only intermediate states for t and not T, while it is equal to that of $\mathcal{D}_T$ in all other states.

Then, $Pr_u^{\mathcal{D}_{cmb}}(\Diamond s) = \overline{Pr}_u^{\mathcal{I}}(\Diamond s)$ for all $u \in T \cup \{t\}$, i.e., $\mathcal{D}_{cmb}$ has the maximum reachability probability for all states in $T \cup \{t\}$.

Conversely, if $Pr_u^{\mathcal{D}_T}(\Diamond s) = \underline{Pr}_u^{\mathcal{I}}(\Diamond s)$ for all $u \in T$ and $Pr_t^{\mathcal{D}_t}(\Diamond s) = \underline{Pr}_t^{\mathcal{I}}(\Diamond s)$, the identical construction of $\mathcal{D}_{cmb}$ implies $Pr_u^{\mathcal{D}_{cmb}}(\Diamond s) = \underline{Pr}_u^{\mathcal{I}}(\Diamond s)$ for all $u \in T \cup \{t\}$.

Intuitively, this lemma states that using two DTMCs $\mathcal{D}_T$ and $\mathcal{D}_t$ which have optimal reachability probabilities from T and t to s, respectively, we can construct a DTMC $\mathcal{D}_{cmb}$ that has optimal reachability from $T \cup \{t\}$ to s.

Proof 13
First, as $\mathcal{D}_t, \mathcal{D}_T \models_U \mathcal{I}$ it follows from the construction that also $\mathcal{D}_{cmb} \models \mathcal{I}$. Next, as for each $w \in S^{\mathcal{I}}$ either $\mathbf{P}^{\mathcal{D}_{cmb}}(w,v) = \mathbf{P}^{\mathcal{D}_t}(w,v)$ or $\mathbf{P}^{\mathcal{D}_{cmb}}(w,v) = \mathbf{P}^{\mathcal{D}_T}(w,v)$ for all $v \in S^{\mathcal{I}}$ it follows that $\mathbf{P}^{\mathcal{D}_{cmb}}$ is well-defined, i.e., a discrete probability distribution, see Def. 2.8.

As $\mathbf{P}^{\mathcal{D}_{cmb}}(w,v) = \mathbf{P}^{\mathcal{D}_T}(w,v)$ for all $w \in I_{T,s}^{\mathcal{D}_T}$, $\overline{Pr}_u^{\mathcal{I}}(\Diamond s) = Pr_u^{\mathcal{D}_{cmb}}(\Diamond s)$ for all $u \in T$, see Lemma 3.7.

Thus, we need to show that $Pr_t^{\mathcal{D}_{cmb}}(\Diamond s) = \overline{Pr}_t^{\mathcal{I}}(\Diamond s)$. As we know that $Pr_t^{\mathcal{D}_t}(\Diamond s) = \overline{Pr}_t^{\mathcal{I}}(\Diamond s)$, we will prove that $Pr_t^{\mathcal{D}_{cmb}}(\Diamond s) = Pr_t^{\mathcal{D}_t}(\Diamond s)$. Notice, that if $I_{t,s}^{\mathcal{D}_t} \cap I_{T,s}^{\mathcal{D}_T} = \emptyset$, then the proof concludes using the same argument as for $u \in T$. Furthermore, if $I_{t,s}^{\mathcal{D}_t} \subseteq I_{T,s}^{\mathcal{D}_T}$, then the proof concludes directly using Theorem 5.8.

Recall, that $Pr_t^{\mathcal{D}_{cmb}}(\Diamond s) = \sum_{u \in N(t) \cap I_{t,s}^{\mathcal{D}_{cmb}}} \mathbf{P}^{\mathcal{D}_{cmb}}(t,u) \cdot Pr_u^{\mathcal{D}_{cmb}}(\Diamond s)$, see Lemma 3.7.
For all $u \in I_{t,s}^{\mathcal{D}_t} \setminus I_{T,s}^{\mathcal{D}_T}$, $\mathbf{P}^{\mathcal{D}_{cmb}}(u,w) = \mathbf{P}^{\mathcal{D}_t}(u,w)$ and for all $u \in I_{T,s}^{\mathcal{D}_T} \cap I_{t,s}^{\mathcal{D}_t}$, $Pr_u^{\mathcal{D}_{cmb}}(\Diamond s) = Pr_u^{\mathcal{D}_t}(\Diamond s)$, see Corollary 5.9. From this, we can conclude that $Pr_t^{\mathcal{D}_t}(\Diamond s) = Pr_t^{\mathcal{D}_{cmb}}(\Diamond s)$. Thus, we have shown that $Pr_u^{\mathcal{D}_{cmb}}(\Diamond s) = \overline{Pr}_u^{\mathcal{D}_{cmb}}(\Diamond s)$ for all $u \in T \cup \{t\}$.

Conversely, the same holds for minimum reachability probabilities using this reasoning, given that $Pr_u^{\mathcal{D}_T}(\Diamond s) = \underline{Pr}_u^{\mathcal{I}}(\Diamond s)$ for all $u \in T$ and $Pr_t^{\mathcal{D}_t}(\Diamond s) = \underline{Pr}_t^{\mathcal{I}}(\Diamond s)$.□

Crucially, Lemma 5.10 can be extended to all states.

Theorem 5.11
Given an IDTMC $\mathcal{I}$ and a state $s \in S^{\mathcal{I}}$ there exists a DTMC $\mathcal{D} \models_U \mathcal{I}$ such that $Pr_t^{\mathcal{D}}(\Diamond s) = \overline{Pr}_t^{\mathcal{I}}(\Diamond s)$ for all states $t \in S^{\mathcal{I}}$.

Proof 14
Consider any given enumeration $t_1, t_2, \ldots$ of $S^{\mathcal{I}}$. Clearly, this enumeration is finite, since $S^{\mathcal{I}}$ is finite, see Assumption 1.

Construction. Now, let $\mathcal{D}^{(1)} \models_U \mathcal{I}$ with $Pr_{t_1}^{\mathcal{D}^{(1)}}(\Diamond s) = \overline{Pr}_{t_1}^{\mathcal{I}}$ and $T_1 \coloneqq I_{t_1,s}^{\mathcal{D}^{(1)}}$, trivially such a DTMC must exist. By Theorem 5.8, $Pr_u(\Diamond s) = \overline{\mathbf{P}}_u(\Diamond s)$.

Inductively, $\mathcal{D}^{(n)} \models_U \mathcal{I}$ (and $s \in T_n$). If $t_{n+1} \in T_n$, then we set $\mathcal{D}^{(n+1)} \coloneqq \mathcal{D}^{(n)}$ and $T_{n+1} = T_n$. If not, we use Lemma 5.10 with $T = T_n$, $\mathcal{D}_T = \mathcal{D}^{(n)}$, and $t = t_{n+1}$ to obtain $\mathcal{D}^{(n+1)} \coloneqq \mathcal{D}_{cmb}$ with $\mathcal{D}^{(n+1)} \models_U \mathcal{I}$. We set $T_{n+1} = T_n \cup I_{t_{n+1},s}^{\mathcal{D}^{(n+1)}}$, thus and $Pr_{t_i}^{\mathcal{D}^{(n+1)}}(\Diamond s) = \overline{Pr}_{t_i}^{\mathcal{I}}(\Diamond s)$ for all $t_i \in T_{n+1}$, as $Pr_{t_i}^{\mathcal{D}^{(n+1)}}(\Diamond s) = \overline{Pr}_{t_i}^{\mathcal{I}}(\Diamond s)$ for all $t_i \in T_n$ (induction), $Pr_{t_{n+1}}^{\mathcal{D}^{(n+1)}}(\Diamond s) = \overline{Pr}_{t_{n+1}}^{\mathcal{I}}(\Diamond s)$ (see Lemma 5.10), and $Pr_u^{\mathcal{D}^{(n+1)}}(\Diamond s) = \overline{Pr}_u^{\mathcal{I}}(\Diamond s)$ for all $u \in I_{t_{n+1},s}^{\mathcal{D}^{(n+1)}}$ (see Theorem 5.8). □

Thus, to conclude, for any given IDTMC $\mathcal{I}$ and state $s \in S^{\mathcal{I}}$, there exist $\mathcal{D}_{max}, \mathcal{D}_{min} \models_U \mathcal{I}$ with maximum and minimum reachability probability, respectively, for all states $t \in S^{\mathcal{I}}$.

Recall, that the ℓEVTs and uEVTs are given by the optimal ratios of Eq. (3.6): $\mathbb{E}^{\mathcal{D}}[vt_s] = \sum_{t \in S_{tr \cap in}} \frac{Pr_t^{\mathcal{D}}(\Diamond s)}{1 - Pr_s^{\mathcal{D}}(\bigcirc \Diamond s)} \cdot \iota_{init}(t)$. Thus, we can conclude the following from the existence of $\mathcal{D}_{max}$ and $\mathcal{D}_{min}$.

Theorem 5.12
Given an IDTMC $\mathcal{I}$ and state $s \in S^{\mathcal{I}}$:

(i) There exists no $\mathcal{D} \models_U \mathcal{I}$ such that $\mathbb{E}^{\mathcal{D}_{max}}[vt_s] < \mathbb{E}^{\mathcal{D}}[vt_s]$.

Conversely, given $\mathcal{D}_{min}$, there exists no $\mathcal{D} \models_U \mathcal{I}$ such that $\mathbb{E}^{\mathcal{D}_{min}}[vt_s] > \mathbb{E}^{\mathcal{D}}[vt_s]$.

In other words $\mathbb{E}^{\mathcal{D}_{max}}[vt_s] = \overline{\mathbb{E}}^{\mathcal{I}}[vt_s]$ and $\mathbb{E}^{\mathcal{D}_{min}}[vt_s] = \underline{\mathbb{E}}^{\mathcal{I}}[vt_s]$ under the UMC semantic.

(ii) The ℓEVT and uEVT of s are both identical under the two semantics, i.e., the semantic does not have any impact on the value of the ℓEVT and uEVT

Proof 15
We start by proving (i).

As we showed in Theorem 5.11, there exist DTMCs $\mathcal{D}_{max}, \mathcal{D}_{min} \models_U \mathcal{I}$ for any given IDTMC $\mathcal{I}$ that have maximum or minimum reachability from all states to s, i.e.,

$$Pr_t^{\mathcal{D}_{max}}(\Diamond s) = \overline{Pr}_t^{\mathcal{I}}(\Diamond s) \qquad \text{or} \qquad Pr_t^{\mathcal{D}_{min}}(\Diamond s) = \underline{Pr}_t^{\mathcal{I}}(\Diamond s)$$

for all $t \in S^{\mathcal{I}}$. We fix these two DTMCs for rest of this proof. Further, we can thus conclude that

$$Pr^{\mathcal{D}_{max}}(\Diamond s) = \overline{Pr}^{\mathcal{I}}(\Diamond s) \qquad \text{and} \qquad Pr_s^{\mathcal{D}_{max}}(\bigcirc\Diamond s) = \overline{Pr}_s^{\mathcal{I}}(\bigcirc\Diamond s) \tag{5.2}$$

or

$$Pr^{\mathcal{D}_{min}}(\Diamond s) = \underline{Pr}^{\mathcal{I}}(\Diamond s) \qquad \text{and} \qquad Pr_s^{\mathcal{D}_{min}}(\bigcirc\Diamond s) = \underline{Pr}_s^{\mathcal{I}}(\bigcirc\Diamond s)\ . \tag{5.3}$$

Using Eq. (3.7), we can thus differentiate the following three cases for s:

1. If $s \in S^{\mathcal{I}}_{\forall re}$, then $\overline{Pr}_s^{\mathcal{I}}(\bigcirc\Diamond s) = \underline{Pr}_s^{\mathcal{I}}(\bigcirc\Diamond s) = 1$ and thus Eq. (3.7) is undefined for $\overline{\mathbb{E}}[vt_s]$ and $\underline{\mathbb{E}}[vt_s]$. However, using Theorem 5.4 we can conclude that $\overline{\mathbb{E}}[vt_s] = \underline{\mathbb{E}}[vt_s] = \infty$. Using Eq. (5.2) and Eq. (5.3), we can deduce that s is reachable and recurrent in both $\mathcal{D}_{max}$ and $\mathcal{D}_{min}$. Thus, using Lemma 3.18, we can conclude that $\mathbb{E}^{\mathcal{D}_{max}}[vt_s] = \overline{\mathbb{E}}[vt_s] = \infty$ and $\mathbb{E}^{\mathcal{D}_{min}}[vt_s] = \underline{\mathbb{E}}[vt_s] = \infty$.

2. If $s \in S^{\mathcal{I}}_{re}$, then $\overline{Pr}_s^{\mathcal{I}}(\bigcirc\Diamond s) = 1$ and thus Eq. (3.7) is undefined for $\overline{\mathbb{E}}[vt_s]$. However, using Theorem 5.4 we can conclude that $\overline{\mathbb{E}}[vt_s] = \infty$. Analogously to case (1), using Eq. (5.2) and Lemma 3.18 it follows that $\mathbb{E}^{\mathcal{D}_{max}}[vt_s] = \overline{\mathbb{E}}[vt_s] = \infty$. For the calculation of $\underline{\mathbb{E}}[vt_s]$, see case (3).

3. In all remaining cases, we can calculate the ℓEVT and uEVT using Eq. (3.7)

$$\underline{\mathbb{E}}[vt_s] = \frac{\underline{Pr}^{\mathcal{I}}(\Diamond s)}{1 - \underline{Pr}_s^{\mathcal{I}}(\bigcirc\Diamond s)} = \frac{Pr^{\mathcal{D}_{min}}(\Diamond s)}{1 - Pr_s^{\mathcal{D}_{min}}(\bigcirc\Diamond s)} = \mathbb{E}^{\mathcal{D}_{min}}[vt_s]$$

and

$$\overline{\mathbb{E}}[vt_s] = \frac{\overline{Pr}^{\mathcal{I}}(\Diamond s)}{1 - \overline{Pr}_s^{\mathcal{I}}(\bigcirc\Diamond s)} = \frac{Pr^{\mathcal{D}_{max}}(\Diamond s)}{1 - Pr_s^{\mathcal{D}_{max}}(\bigcirc\Diamond s)} = \mathbb{E}^{\mathcal{D}_{max}}[vt_s]\ .$$

We continue by shortly proving the equations presented in case (3) by contradiction. Let us assume that there exists a DTMC $\mathcal{D} \models \mathcal{I}$ such that $\mathbb{E}^{\mathcal{D}_{max}}[vt_s] < \mathbb{E}^{\mathcal{D}}[vt_s]$. Using Eq. (3.6), it thus follows that either

$$Pr^{\mathcal{D}}(\Diamond s) > Pr^{\mathcal{D}_{max}}(\Diamond s) \qquad \text{or} \qquad Pr^{\mathcal{D}}(\bigcirc\Diamond s) > Pr^{\mathcal{D}_{max}}(\bigcirc\Diamond s)\ . \qquad \square$$

However, as $Pr^{\mathcal{D}_{max}}(\Diamond s) = \overline{Pr}^{\mathcal{I}}(\Diamond s)$ and $Pr^{\mathcal{D}_{max}}(\bigcirc\Diamond s) = \overline{Pr}^{\mathcal{I}}(\bigcirc\Diamond s)$ this leads to a contradiciton. The proof for $\mathcal{D}_{min}$ follows analogously by reversing the inequalities.

Lastly, we prove (ii). Recall, that for any DTMC $\mathcal{D} \models_U \mathcal{I}$, then $\mathcal{D} \models_I \mathcal{I}$, see Theorem 4.15. Thus, $\mathcal{D}_{max}, \mathcal{D}_{min} \models_I \mathcal{I}$. Let us assume that $\mathcal{D}_{max}$ and $\mathcal{D}_{min}$ are induced by some stationary natures η_{max} and η_{min}. Further, let us assume that $\mathcal{D}_{\eta_{max}} \models \mathcal{I}$ and $\mathcal{D}_{\eta_{min}}$ are the infinite-state representations of η_{max} and η_{min}, see Def. 4.7.

As stated in [MKQW25], the EVTs of states $s \in S$ in acyclic DTMCs $\mathcal{D}$ coincide with the reachability probability of s, i.e., $\mathbb{E}[vt_s] = Pr(\Diamond s)$. Now recall, that the natures η on $\mathcal{I}$ can be represented by acyclic infinite-state DTMCs $\mathcal{D}_\eta$, see Def. 4.7. Thus, the expected visiting time of the states $t \in \tau^{-1}(T)$ is given by, $\mathbb{E}^{\mathcal{D}_\eta}[vt_t] = Pr^{\mathcal{D}_\eta}(\Diamond t)$. Consequently, $\underline{\mathbb{E}}^{\mathcal{I}}[vt_s] = \inf_{\mathcal{D}\models_I\mathcal{I}} Pr^{\mathcal{D}}(\bigcup_{t\in\tau^{-1}(s)} \Diamond t)$ and $\overline{\mathbb{E}}^{\mathcal{I}}[vt_s] = \sup_{\mathcal{D}\models_I\mathcal{I}} Pr^{\mathcal{D}}(\bigcup_{t\in\tau^{-1}(s)} \Diamond t)$, see Def. 5.1. !!!

5.2.1 UMC semantic

In this section we examine how the precise values of the uEVTs and lEVTs can be calculated for IDTMCs under the UMC semantic.

As defined, there can be multiple DTMCs $\mathcal{D}$ that satisfy an IDTMC $\mathcal{I}$ with differing transition probability functions $\mathbf{P}^{\mathcal{D}}$ and initial distributions $\iota^{\mathcal{D}}_{init}$. To determine the uEVT and ℓEVT for a given state $s \in S^{\mathcal{I}}$, we thus have to find the satisfying DTMCs that maximize and minimze Eq. (3.2), respectively. In other words, we need to solve the equations

$$\mathbb{E}[vt_s] = \text{opt}_{\mathcal{D}\models\mathcal{I}} \left(\iota_{init}(s) + \sum_{t\in S_{tr}} \mathbf{P}^{\mathcal{D}}(t,s) \cdot \mathbb{E}[vt_t] \right), \qquad \text{opt} \in \{\min, \max\}.$$

In [CHK13], the authors construct the following NLP (which we have rewritten slightly) to compute any maximum or minimum reachability, respectively, for a set of nodes $T \subseteq S$, using the fixed-initial-state-definition.

$$\begin{aligned}
\text{opt} \quad & \sum_{s\in S\setminus T} x_s && && \text{opt} \in \{\max, \min\} \\
\text{s.t.} \quad & x_s \sim \sum_{s'\in S\setminus T} \mathbf{P}(s,s')x_{s'} + \sum_{s'\in T} \mathbf{P}(s,s') && \forall s \in S\setminus T \text{ and } \sim\in\{\leq,\geq\} && (1)\\
& x_s = 1 && \forall s \in T && (2)\\
& \sum_{s_j\in N(s)} \mathbf{P}(s_i,s_j) = 1 && \forall s_i \in S_{tr} && (3)\\
& \underline{\mathbf{P}}^{\mathcal{I}}(s_i,s_j) \leq \mathbf{P}(s_i,s_j) \leq \overline{\mathbf{P}}^{\mathcal{I}}(s_i,s_j) && \forall s_i,s_j \in S^{\mathcal{I}}_{tr} && (4)
\end{aligned}$$

(Allowed only max + $\leq$ and min + $\geq$)

As shown in [CHK13], this NLP can be solved in PTIME. Thus given an IDTMC $\mathcal{I}$ and state $s \in S^{\mathcal{I}}$, we can calculate the maximum or minimum reachability probabilities $\overline{Pr}^{\mathcal{I}}(\Diamond s)$ and $\overline{Pr}^{\mathcal{I}}(\bigcirc\Diamond s)$ or $\underline{Pr}^{\mathcal{I}}(\Diamond s)$ and $\underline{Pr}^{\mathcal{I}}(\bigcirc\Diamond s)$ of a state s in PTIME.

Recall, that the ℓEVT and uEVT of s can be calculated through a polynomial function (fraction) for the UMC semantic using only $\overline{Pr}^{\mathcal{I}}(\Diamond s)$ and $\overline{Pr}^{\mathcal{I}}(\bigcirc\Diamond s)$ or $\underline{Pr}^{\mathcal{I}}(\Diamond s)$ and $\underline{Pr}^{\mathcal{I}}(\bigcirc\Diamond s)$, see Theorem 5.12.

Thus, we can calculate the ℓEVT and uEVT of s in PTIME for the UMC semantic.

5.2.2 IMDP semantic

In this subsection we will look at how uEVTs and lEVTs can be calculated for the IMDP semantic.

By combining our conclusion that the ℓEVT and uEVT of a given state $s \in S^{\mathcal{I}}$ under the UMC semantic can be calculated in PTIME and claim (ii) of Theorem 5.12, we can intuitively conclude that the ℓEVT and uEVT of s under the IMDP semantic can be calculated in PTIME as well.

In the following, we present a different approach to prove this result. By showing this approach both the relation between the reachability probabilities and expected visiting times as well as the differences between the UMC and IMDP semantic becom more apparant. Moreover, this result allows us to infer some additional results for the IMDP semantic.

First, we define the DTMCs

$$\mathcal{D}_{s_{in}\to s} = (S^{\mathcal{I}}, \mathbf{P}^{\mathcal{D}_{s_{in}\to s}}, \iota_{init}^{\mathcal{D}_{s_{in}\to s}}) \qquad \text{and} \qquad \mathcal{D}_{s\to s} = (S^{\mathcal{I}}, \mathbf{P}^{\mathcal{D}_{s\to s}}, \iota_{init}^{\mathcal{D}_{s\to s}})$$

such that $\mathcal{D}_{s_{in}\to s}, \mathcal{D}_{s\to s} \models_U \mathcal{I}$ and

$$Pr^{\mathcal{D}_{s_{in}\to s}}(\Diamond s) = \overline{Pr}^{\mathcal{I}}(\Diamond s) \qquad \text{and} \qquad Pr_s^{\mathcal{D}_{s\to s}}(\bigcirc\Diamond s) = \overline{Pr}_s^{\mathcal{I}}(\bigcirc\Diamond s)\ . \tag{5.4}$$

Recall, that a DTMC $\mathcal{D} \models_I \mathcal{I}$ satisfies an IDTMC $\mathcal{I}$ if and only if it is induced by a nature η_{aug} on $\mathcal{I}$. Notice, that we can define a nature that chooses the probability distribution specified by $\mathcal{D}_{s_{in}\to s}$ until state s is visited for the first time and then chooses the probability distribution specified by $\mathcal{D}_{s\to s}$. Formally, we can define this nature as follows:

$$\eta_{aug} : Paths_{fin}^{\mathcal{I}} \to Steps^{\mathcal{I}}$$

such that

$$\eta_{aug}(\pi) = \begin{cases} \mathbf{P}^{\mathcal{D}_{s_{in}\to s}}(last(\pi)) & \text{if } \nexists k \in \underline{|\pi|} \text{ with } \pi[k] = s \\ \mathbf{P}^{\mathcal{D}_{s\to s}}(last(\pi)) & \text{otherwise.} \end{cases}$$

This nature can be characterized by having two stages where we transition from the first to the second stage when passing through s for the first time.

We continue by defining the DTMC $\mathcal{D}^s_{aug}$ using $\mathbf{P}^{\mathcal{D}_{s_{in}\to s}}$ and $\mathbf{P}^{\mathcal{D}_{s\to s}}$.

Construction. Given an IDTMC $\mathcal{I} = (S, \underline{\mathbf{P}}, \overline{\mathbf{P}}, \underline{\iota}_{init}, \overline{\iota}_{init})$, the previously defined $\mathcal{D}_{s_{in}\to s}$, $\mathcal{D}_{s\to s}$ (corresponding to either the suprema or infima) and a state s (for which we want to calculate the EVT), we define $\mathcal{D}^s_{aug}$, with $S^{\mathcal{D}^s_{aug}} = S_{st1} \cup S_{st2}$, where the state set for stage 1 is

$$S_{st1} = \{s^{(1)} \mid s \in S^{\mathcal{I}} = S^{\mathcal{D}_{s_{in}\to s}}\}$$

and the state set for stage 2 is

$$S_{st2} = \{s^{(2)} \mid s \in S^{\mathcal{I}} = S^{\mathcal{D}_{s\to s}}\} .$$

Furthermore,

$$\mathbf{P}^{\mathcal{D}^s_{aug}}(s_1^{(1)}, s_2^{(1)}) = \mathbf{P}^{\mathcal{D}_{s_{in}\to s}}(s_1, s_2)$$

for all $s_1^{(1)} \in S_{st1}\backslash\{s\}$, $s_2^{(1)} \in S_{st1}$ and $\mathbf{P}^{\mathcal{D}^s_{aug}}(s^{(1)}, s^{(2)}) = 1$. Thus, $\mathcal{D}^s_{aug}$ has the transition probability function of $\mathcal{D}_{s_{in}\to s}$ until s is reached for the first time and $\mathcal{D}^s_{aug}$ necessariliy transitions from the stage 1 copy of s, namely $s^{(1)}$, to the stage 2 copy of s, namely $s^{(2)}$.

Similarly,

$$\mathbf{P}^{\mathcal{D}^s_{aug}}(s_1^{(2)}, s_2^{(2)}) = \mathbf{P}^{\mathcal{D}_{s\to s}}(s_1, s_2)$$

for all $s_1^{(2)}, s_2^{(2)} \in S_{st2}$, i.e., $\mathcal{D}^s_{aug}$ has the transition probability function of $\mathcal{D}_{s\to s}$ after s has been reached for the first time. All other pairs of states are mapped to 0 by $\mathbf{P}^{\mathcal{D}^s_{aug}}$. Thus, $\mathcal{D}^s_{aug}$ cannot transition back to states of stage 1, after s has been entered.

Lastly,

$$\iota_{init}^{\mathcal{D}^s_{aug}}(t^{(1)}) = \iota_{init}^{\mathcal{D}_{s_{in}\to s}}(t)$$

for all $t^{(1)} \in S_{st1} = S^{\mathcal{D}_{s_{in}\to s}}$. All other states are mapped to 0 by $\iota_{init}^{\mathcal{D}^s_{aug}}$

To illustrate this construction of $\mathcal{D}^s_{aug}$, we provide the following example.

Example 5.13
Recall the IDTMC $\mathcal{I}$ in Fig. 4.1, displayed again in Fig. 5.3. Fig. 5.4 shows $\mathcal{D}^{s_4}_{aug}$, i.e., the construction of the maximizing DTMC for the EVT of $\mathcal{I}$.

We now show that $\mathcal{D}^s_{aug}$ satisfies $\mathcal{I}$ under the IMDP semantic.

Lemma 5.14
Given and IDTMC $\mathcal{I}$, a state s, and the corresponding $\mathcal{D}^s_{aug}$, it holds that $\mathcal{D}^s_{aug} \models_I \mathcal{I}$

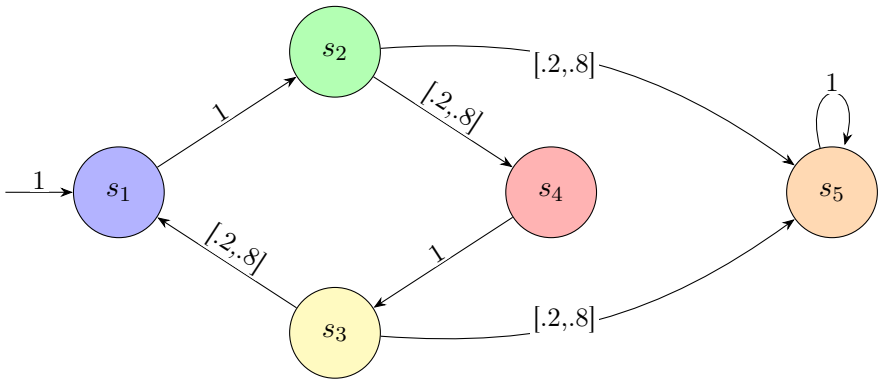


Figure 5.3: target state in red

Proof 16
We show that $\mathcal{D}^s_{aug}$ is induced by the nature η_{aug} on $\mathcal{I}$ defined by

$$\eta_{aug}(\pi) = \begin{cases} \mathbf{P}^{\mathcal{D}_{s_{in}\to s}}(last(\pi)) & \text{if } \nexists k \in \underline{|\pi|} \text{ with } \pi[k] = s \\ \mathbf{P}^{\mathcal{D}_{s\to s}}(last(\pi)) & \text{otherwise.} \end{cases}$$

Since $\mathcal{D}_{s_{in}\to s} \models_U \mathcal{I}$ and $\mathcal{D}_{s\to s} \models_U \mathcal{I}$, both transition distributions are admissible in $\mathcal{I}$. Hence η_{aug} is a valid nature on $\mathcal{I}$.

Before the first visit to s, η_{aug} chooses the transition probabilities of $\mathcal{D}_{s_{in}\to s}$. Therefore, for all $s_i, s_j \in S$ with $s_i \neq s$,

$$\mathbf{P}^{\mathcal{D}^s_{aug}}(s_i^{(1)}, s_j^{(1)}) = \mathbf{P}^{\mathcal{D}_{s_{in}\to s}}(s_i, s_j) = \eta_{\text{aug}}(\pi)(s_j)$$

for every finite path π ending in s_j that does not contain s. This equation holds by the construction of $\mathcal{D}^s_{aug}$ and η_{aug}. If the transition reaches s, the augmented DTMC enters the second stage.

After the first visit to s, η_{aug} chooses the transition probabilities of $\mathcal{D}_{s\to s}$. Consequently, for all $s_i, s_j \in S$,

$$\mathbf{P}^{\mathcal{D}^s_{aug}}(s_i^{(2)}, s_j^{(2)}) = \mathbf{P}^{\mathcal{D}_{s\to s}}(s_i, s_j) = \eta_{\text{aug}}(\pi)(s_j)$$

for every finite path π ending in t that contains s. This equation holds by the construction of $\mathcal{D}^s_{aug}$ and η_{aug}.

Finally, the initial distribution of $\mathcal{D}^s_{aug}$ is obtained from the initial distribution of $\mathcal{D}_{s_{in}\to s}$ by assigning it to the first stage copies of the states for all $t \in S_{tr}^{\mathcal{D}_{s_{in}\to s}}$. Hence, the initial distribution agrees with that induced by η_{aug}.

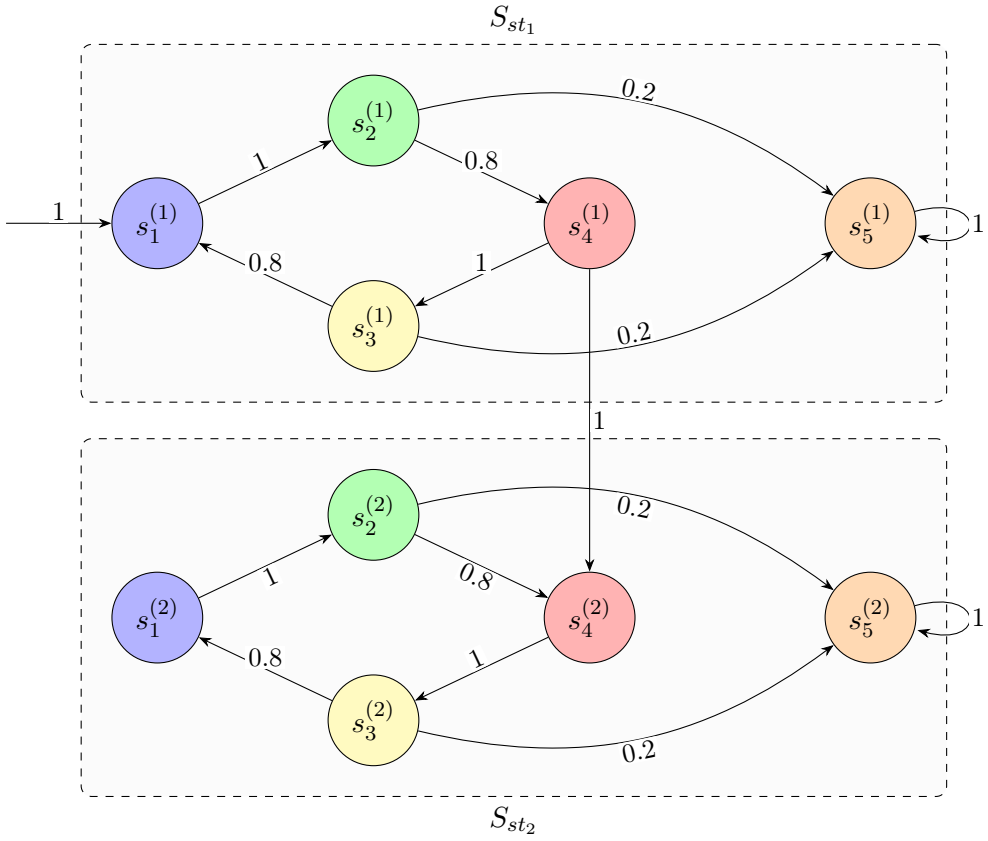


Figure 5.4: $\tau(t_1) = \tau(t_4) = s_1$, $\tau(t_2) = \tau(t_7) = s_2$, $\tau(t_5) = s_3$, $\tau(t_3) = \tau(t_6) = s_4$. Target state in red.

Thus, $\mathcal{D}^s_{aug}$ is precisely the DTMC induced by η_{aug} on $\mathcal{I}$. Therefore,

$$\mathcal{D}^s_{aug} \models_I \mathcal{I}. \qquad \square$$

Using Eq. (5.4), we can conclude that

$$Pr^{\mathcal{D}^s_{aug}}(\Diamond s) = \overline{Pr}^{\mathcal{I}}(\Diamond s) \qquad \text{and} \qquad Pr_s^{\mathcal{D}^s_{aug}}(\bigcirc\Diamond s) = \overline{Pr}_s^{\mathcal{I}}(\bigcirc\Diamond s)\ .$$

Thus, using the same case distinction as in Proof 15 for Eq. (3.7), we can conclude that, similar to Theorem 5.12, $\mathbb{E}^{\mathcal{D}^s_{aug}}[vt_s] = \overline{\mathbb{E}}^{\mathcal{I}}[vt_s]$.

Conversely, if we define

$$Pr^{\mathcal{D}_{s_{in}\to s}}(\Diamond s) = \underline{Pr}^{\mathcal{I}}(\Diamond s) \qquad \text{and} \qquad Pr_s^{\mathcal{D}_{s\to s}}(\bigcirc\Diamond s) = \underline{Pr}_s^{\mathcal{I}}(\bigcirc\Diamond s)$$

and construct η_{aug} and $\mathcal{D}^s_{aug}$, as described in this section, then $\mathbb{E}^{\mathcal{D}^s_{aug}}[vt_s] = \underline{\mathbb{E}}^{\mathcal{I}}[vt_s]$ using the same reasoning.

Intuitively, η_{aug} is not a stationary nature, as $\eta_{aug} : Paths^{\mathcal{I}}_{fin} \to Steps^{\mathcal{I}}$ depends not only on the current state of the model. Thus, we can conclude that there exists a nature η_{aug}, which is not stationary, for every state $s \in S^{\mathcal{I}}$, that induces a DTMC $\mathcal{D}^s_{aug}$ such that the EVT of s in $\mathcal{D}^s_{aug}$ is equal to the optimal EVT, i.e., ℓEVT or uEVT, of s.

5.3 P-completeness

In this section, we further examine the complexity of calculating ℓEVTs and uEVTs by showing that the decision problem of ℓEVTs and uEVTs is P-complete.

As shown in [CHK13], computing the maximum or minimum reachability probability is P-complete for the IMDP semantic, and thus also for the UMC semantic.

Recall that computing ℓEVTs and uEVTs is in PTIME, see Sec. 5.2.1. We now show that computing ℓEVTs and uEVTs is P-complete, by reducing it to computating minimum and maximum reachability probabilities, which is P-complete. As both ℓEVTs and uEVTs as well as minimum and maximum rachability probabilities coincide for the two semantics, it suffices to prove the following reduction for the UMC semantic.

First, we introduce the decision problem of ℓEVTs and uEVTs as well as minimum and maximum reachability probabilities.

Definition 5.15 (Decision Problem of ℓEVTs and uEVTs)
Given an IDTMC $\mathcal{I}$, a state $s \in S$, and $\underline{x} \in \mathbb{R}_{>0}$, decide whether $\underline{\mathbb{E}}^{\mathcal{I}}[vt_s] = \underline{x}$

Similarly, given $\overline{x} \in \mathbb{R}_{>0}$, decide whether $\overline{\mathbb{E}}^{\mathcal{I}}[vt_s] = \overline{x}$.

Definition 5.16 (Decision Problem of optimal reachability probabilities)
Given an IDTMC $\mathcal{I}$, a state $s \in S$, and $\underline{x} \in \mathbb{R}_{>0}$, decide whether $\underline{Pr}^{\mathcal{I}}[vt_s] = \underline{x}$

Similarly, given $\overline{x} \in \mathbb{R}_{>0}$, decide whether $\overline{Pr}^{\mathcal{I}}[vt_s] = \overline{x}$.

Thus, we can now formulate the following theorem.

Theorem 5.17
The decision problem of ℓEVTs and uEVTs is P-complete.

Reduction. For the rest of this section, we fix an IDTMC $\mathcal{I}$, a set of states $T \subseteq S^{\mathcal{I}}$ and a state $s \notin S^{\mathcal{I}}$. We then construct the IDTMC $\mathcal{I}_{red} = (S^{\mathcal{I}} \cup \{s, \bot\}, \underline{\mathbf{P}}^{\mathcal{I}_{red}}, \overline{\mathbf{P}}^{\mathcal{I}_{red}}, \iota^{\mathcal{I}}_{init}, \bar{\iota}^{\mathcal{I}}_{init})$. Furthermore, we define $\overline{\mathbf{P}}^{\mathcal{I}_{red}}(u, w) = \overline{\mathbf{P}}^{\mathcal{I}}(u, w)$ and $\underline{\mathbf{P}}^{\mathcal{I}_{red}}(u, w) = \underline{\mathbf{P}}^{\mathcal{I}}(u, w)$ for all $u \in S^{\mathcal{I}} \backslash T$ and $w \in S^{\mathcal{I}}$. Moreover, $\overline{\mathbf{P}}^{\mathcal{I}_{red}}(u, w) = \underline{\mathbf{P}}^{\mathcal{I}_{red}}(u, w) = 0$ and $\overline{\mathbf{P}}^{\mathcal{I}_{red}}(u, s) = \underline{\mathbf{P}}^{\mathcal{I}_{red}}(u, s) = 1$ for $u \in T$ and $w \in S^{\mathcal{I}}$. Finally, $\overline{\mathbf{P}}^{\mathcal{I}_{red}}(s, \bot) = \underline{\mathbf{P}}^{\mathcal{I}_{red}}(s, \bot) = 1$ and $\overline{\mathbf{P}}^{\mathcal{I}_{red}}(\bot, \bot) = \underline{\mathbf{P}}^{\mathcal{I}_{red}}(\bot, \bot) = 1$.

Remark. Intuitively, the mapping $(\mathcal{I}, T) \mapsto (\mathcal{I}_{red})$ is computable in logarithmic space.

Theorem 5.18
Let $\mathcal{D}_{red} \models_U \mathcal{I}_{red}$ denote a DTMC with

$$\mathbb{E}^{\mathcal{D}_{red}}[vt_s] = \overline{\mathbb{E}}^{\mathcal{I}_{red}}[vt_s]$$

and $\mathcal{D}_{max} \models_U \mathcal{I}$ denote a DTMC with

$$Pr^{\mathcal{D}_{max}}(\Diamond T) = \overline{Pr}^{\mathcal{I}}(\Diamond T) \ .$$

Then, $\mathbb{E}^{\mathcal{D}_{red}}[vt_s] = Pr^{\mathcal{D}_{max}}(\Diamond T)$.

Notice, that these optima exist, see the first remark in Sec. 5.2 and Def. 5.1.

In other words, the optimal EVT for s in $\mathcal{I}_{red}$ is equal to the optimal reachability probability of T in $\mathcal{I}$. We prove this theorem by showing that the (maximum) reachability probabilities of T and s are identical in $\mathcal{I}_{red}$ and $\mathcal{D}_{red}$. Then, we verify that the uEVT of s is equal to the maximum reachability probability of s in $\mathcal{I}_{red}$ and the EVT of s is equal to the reachability probability of s in $\mathcal{D}_{red}$. Lastly, we show that the reachability probability of T is identical in $\mathcal{D}_{red}$ and $\mathcal{D}_{max}$. By combining these lemmata we then finally prove the theorem.

Lemma 5.19
It holds that $\overline{Pr}^{\mathcal{I}_{red}}(\Diamond T) = \overline{Pr}^{\mathcal{I}_{red}}(\Diamond s)$ and $Pr^{\mathcal{D}_{red}}(\Diamond T) = Pr^{\mathcal{D}_{red}}(\Diamond s)$.

In other words, the maximum reachability probabilities of T and s are identical in $\mathcal{I}_{red}$.

Proof 17
As $\mathbf{P}^{\mathcal{D}}(t, s) > 0$ if and only if $t \in T$ for all $\mathcal{D} \models_U \mathcal{I}_{red}$, it follows from Lemma 3.7 that $Pr^{\mathcal{D}}(\Diamond s) = \sum_{t \in T} \mathbf{P}(t, s) \cdot Pr^{\mathcal{D}}(\Diamond T)$. Because, $\mathbf{P}^{\mathcal{D}}(t, s) = 1$ for all $t \in T$, we can conclude that $Pr^{\mathcal{D}}(\Diamond T) = Pr^{\mathcal{D}}(\Diamond s)$ for all $\mathcal{D} \models_U \mathcal{I}_{red}$. □

Lemma 5.20
It holds that $\overline{\mathbb{E}}^{\mathcal{I}_{red}}[vt_s] = \overline{Pr}^{\mathcal{I}_{red}}(\Diamond s)$ and $\mathbb{E}^{\mathcal{D}_{red}}[vt_s] = Pr^{\mathcal{D}_{red}}(\Diamond s)$.

In other words, the uEVT of $s \in S^{\mathcal{I}_{red}}$ is equal to the maximum reachability probability of s in $\mathcal{I}_{red}$.

Proof 18
$\mathbb{E}^{\mathcal{D}}[vt_s] = Pr(\Diamond s) \cdot \mathbb{E}^{\mathcal{D}}_s[vt_s]$ for all $\mathcal{D} \models_U \mathcal{I}_{red}$, see Lemma 3.21. As, $\overline{Pr}^{\mathcal{I}_{red}}(s, t) > 0$ if and only if $t = \tau$ and τ is absorbing, it follows that $\mathbb{E}^{\mathcal{D}}_s[vt_s] = 1$, as $Pr^{\mathcal{D}}_s(\bigcirc \Diamond s) = 0$, which concludes the proof. □

Lemma 5.21
It holds that $Pr^{\mathcal{D}_{red}}(\Diamond T) = Pr^{\mathcal{D}_{max}}(\Diamond T)$.

In other words, the reachability probability of T is identical in $\mathcal{D}_{red}$ and $\mathcal{D}_{max}$.

Proof 19
Firstly, if $\mathbb{E}^{\mathcal{D}_{red}}[vt_s] = \overline{\mathbb{E}}^{\mathcal{I}_{red}}[vt_s]$ (as defined in the construction), then $Pr^{\mathcal{D}_{red}}(\Diamond s) = \overline{Pr}^{\mathcal{I}_{red}}(\Diamond s)$, see Lemma 5.20. Moreover, using Lemma 5.19, we can then conclude that $Pr^{\mathcal{D}_{red}}(\Diamond T) = \overline{Pr}^{\mathcal{I}_{red}}(\Diamond T)$.

As $Pr^{\mathcal{D}_{red}}(\Diamond T)$ and $Pr^{\mathcal{D}_{max}}(\Diamond T)$ are only dependent on $\mathbf{P}^{\mathcal{D}_{red}}(u,w)$ and $\mathbf{P}^{\mathcal{D}}(u,w)$ with $u \in S^{\mathcal{I}} \backslash T$ respectively, see Lemma 3.7, and $\overline{\mathbf{P}}^{\mathcal{I}_{red}}(u,w) = \overline{\mathbf{P}}^{\mathcal{I}}(u,w)$ as well as $\underline{\mathbf{P}}^{\mathcal{I}_{red}}(u,w) = \underline{\mathbf{P}}^{\mathcal{I}}(u,w)$ for all $u \in S^{\mathcal{I}} \backslash T$ and $w \in S^{\mathcal{I}}$, we can conclude that $\overline{Pr}^{\mathcal{I}}(\Diamond T) = \overline{Pr}^{\mathcal{I}_{red}}(\Diamond T)$. By the definition of our construction, $Pr^{\mathcal{D}_{max}}(\Diamond T) = \overline{Pr}^{\mathcal{I}}(\Diamond T)$ and $Pr^{\mathcal{D}_{red}}(\Diamond T) = \overline{Pr}^{\mathcal{I}_{red}}(\Diamond T)$ (see first paragraph of this proof) which concludes the proof. □

Finally, we can now prove Theorem 5.18.

Proof 20
We simply need to combine the results of the lemmata. According to Lemma 5.21 $Pr^{\mathcal{D}_{max}}(\Diamond T) = Pr^{\mathcal{D}_{red}}(\Diamond T)$. Furthermore, using Lemma 5.19, $Pr^{\mathcal{D}_{red}}(\Diamond T) = Pr^{\mathcal{D}_{red}}(\Diamond s)$. Finally, Lemma 5.20 shows that $Pr^{\mathcal{D}_{red}}(\Diamond s) = \mathbb{E}^{\mathcal{D}_{red}}[vt_s]$. Thus, combining these results $Pr^{\mathcal{D}_{max}}(\Diamond T) = \mathbb{E}^{\mathcal{D}_{red}}[vt_s]$. By definition $\mathbb{E}^{\mathcal{D}_{red}}[vt_s] = \overline{\mathbb{E}}^{\mathcal{I}_{red}}[vt_s]$ and $Pr^{\mathcal{D}_{max}}(\Diamond T) = \overline{Pr}^{\mathcal{I}}(\Diamond T)$. Therefore, the given maximum reachability probability $\overline{Pr}^{\mathcal{I}}(\Diamond T)$ of any IDTMC $\mathcal{I}$ can be determined through calculating uEVTs.□

Conversely, the same holds using this reasoning for the ℓEVTs.

Thus, as we can calculate the reachabilty minimum and maximum reachability probabilities of any given $T \subseteq S^{\mathcal{I}}$ using ℓEVTs and uEVTs, we can decide the problem of optimal reachability probabilities using $\underline{x} = \underline{\mathbb{E}}^{\mathcal{I}}_{red}[vt_s]$ and $\overline{x} = \overline{\mathbb{E}}^{\mathcal{I}}_{red}[vt_s]$. As the decision problem of optimal reachability probabilities is P-complete, it thus follows that the decision problem of ℓEVTs and uEVTs is P-complete as well.

Chapter 6

Stationary Distributions in IDTMCs

In this chapter, we examine how ℓEVTs and uEVTs can be used to calculate stationary distributions in IDTMCs. For this we start by .... To this end, we fix an IDTMC $\mathcal{I} = (S^{\mathcal{I}}, \underline{\mathbf{P}}, \overline{\mathbf{P}}, \underline{\iota}_{init}, \overline{\iota}_{init})$ and a DTMC $\mathcal{D} = (S^{\mathcal{D}}, \mathbf{P}, \iota_{init})$ with $\mathcal{D} \models_U \mathcal{I}$.

We define stationary distributions, also called long-run distributions, as done in [BK08].

Definition 6.1 (Stationary Distributions in DTMCs)
Given a finite DTMC $\mathcal{D} = (S, \mathbf{P}, \iota_{init})$ be a finite Markov chain and $s \in S$. The *long-run average probability* $\theta^{\mathcal{D}}(s)$ is given by

$$\theta^{\mathcal{D}}(s) = \lim_{n \to \infty} \frac{1}{n} \cdot \sum_{i=1}^{n} Pr(\Diamond^{=k} s) \ .$$

In other words, the stationary distribution $\Theta^{\mathcal{D}}(t)$ describes the probability that $\mathcal{D}$ is in state t after k steps, as k tends towards infinity. Whenever the DTMC $\mathcal{D}$ is evident from the context, we omit the superscript $\mathcal{D}$ from $\theta^{\mathcal{D}}$.

Similarly, we now define stationary distributions for IDTMCs. First, recall that the DTMCs $\mathcal{D}_\eta \models_I \mathcal{I}$ that are induced by some nature η can be infinite, i.e., $|\mathcal{D}_\eta| = \infty$, see Def. 4.7. Intuitively, as every state $s \in S^{\mathcal{D}_{eta}}$ in these DTMCs is only visited once, $\theta(s) = 0$. In other words, there exists no stationary distribution for these states. Thus, we restrict this examination of BSCCs, in this thesis, to the UMC semantic.

Definition 6.2 (Stationary Distributions in IDTMCs)
For a given IDTMC $\mathcal{I}$ and state $s \in S^{\mathcal{I}}$, we define

$$\Theta(s) = \{\theta(s)^{\mathcal{D}} \mid \mathcal{D} \models_U \mathcal{I}\}$$

as the set of stationary distributions of s on $\mathcal{I}$.
Furthermore, we define $\underline{\theta}^{\mathcal{I}}(s)$ and $\overline{\theta}^{\mathcal{I}}(s)$ such that

$$\underline{\theta}^{\mathcal{I}}(s) = \inf_{\mathcal{D} \models_U \mathcal{I}} \theta^{\mathcal{D}}(s) \qquad \text{and} \qquad \overline{\theta}^{\mathcal{I}}(s) = \sup_{\mathcal{D} \models_U \mathcal{I}} \theta^{\mathcal{D}}(s) \ .$$

In other words, $\underline{\theta}^{\mathcal{I}}(s)$ and $\overline{\theta}^{\mathcal{I}}(s)$ describe the minimum and maximum stationary distribution of s among the DTMCs $\mathcal{D} \models_U \mathcal{I}$ that satisfy $\mathcal{I}$ under the UMC semantic. Whenever the DTMC $\mathcal{D}$ is evident from the context, we omit the superscript $\mathcal{I}$ from $\underline{\theta}^{\mathcal{I}}$ and $\overline{\theta}^{\mathcal{I}}$.

Moreover, BSCCs are defined as follows for IDTMCs.

Definition 6.3 (BSCCs for IDTMCs)
Let $\mathcal{I}$ be an IDTMC, then $B^{\mathcal{I}} \subseteq S^{\mathcal{I}}$ is a BSCC, if there exists at least one DTMC $\mathcal{D} \models_U \mathcal{I}$ such that $B^{\mathcal{I}}$ is a BSCC in the DTMC $\mathcal{D}$, see Def. 3.9.

Whenever the DTMC $\mathcal{D}$ is evident from the context, we omit the superscript $\mathcal{I}$ from $B^{\mathcal{I}}$.

In contrast to EVTs, when considering the stationary distribution of a DTMC or IDTMC, we are interested in the recurrent, not transient, states.

Recall, that the recurrent states of a DTMC $\mathcal{D}$ are precisely the states contained in the BSCCs of $\mathcal{D}$, see Sec. 3.1. Thus, by collapsing each BSCCS of $\mathcal{D}$ into a single state, the stationary distribution $\theta^{\mathcal{D}}$ of the recurrent states of a DTMC $\mathcal{D}$ is characterized by

$$\theta^{\mathcal{D}}(s) = Pr^{\mathcal{D}}(\Diamond B) \cdot \theta^{\mathcal{D}|_B}(s) \ , \tag{6.1}$$

for all $s \in B$, as proven in [Kul16], where $\theta^{\mathcal{D}|_B}$ denotes the stationary distribution of $\mathcal{D}$ restricted to B. Moreover, in [MKQW25], it is shown that the stationary distribution within a BSCC can be computed through its EVTs. For this they construct a DTMC $\mathcal{B}\,{}^{\hat{v}}$ for every BSCC B with any arbitrary state $v \in S^{\mathcal{D}|_B}$ in which all incoming transitions of v are redirected to the fresh absorbing state $\hat{v}$. As v is arbitrary, we fix v and $\hat{v}$ for the remainder of this chapter.

Construction. More precisely, $\mathcal{B}\,{}^{\hat{v}} = (S^B \cup \{\hat{v}\}, \mathbf{P}^{\mathcal{B}^{\hat{v}}}, \iota_{init}^{\mathcal{B}^{\hat{v}}})$, where S^B denotes the states that are contained in B. Moreover, $\mathbf{P}^{\mathcal{B}^{\hat{v}}}(\hat{v}, \hat{v}) = 1$, $\mathbf{P}^{\mathcal{B}^{\hat{v}}}(s, \hat{v}) = \mathbf{P}^{\mathcal{D}}(s, v)$, and $\mathbf{P}^{\mathcal{B}^{\hat{v}}}(s, v) = 0$ for all $s \in S^{\mathcal{D}|_B}$. Lastly, $\mathbf{P}^{\mathcal{B}^{\hat{v}}}(s, t) = \mathbf{P}^{\mathcal{D}}(s, t)$ for all $s \in S$, $t \in S \backslash \{v\}$, and $\iota_{init}^{\mathcal{B}^{\hat{v}}}(v) = 1$.

The stationary distribution within a DTMC is then given by

$$\theta^{\mathcal{D}|_B}(s) = \frac{\mathbb{E}^{\mathcal{B}\Rsh\hat{v}}[vt_s]}{\sum_{t\in S}\mathbb{E}^{\mathcal{B}\Rsh\hat{v}}[vt_t]}\ , \tag{6.2}$$

as proven in [MKQW25].

Similarly to the DTMC $\mathcal{B}\Rsh\hat{v}$, we now define the IDTMC $\mathcal{I}\Rsh\hat{v}$ as follows.

Construction. $\mathcal{I}\Rsh\hat{v} = (S^B \cup \{\hat{v}\}, \underline{\mathbf{P}}^{\mathcal{B}\Rsh\hat{v}}, \overline{\mathbf{P}}^{\mathcal{B}\Rsh\hat{v}}, \iota_{init}^{\mathcal{B}\Rsh\hat{v}}, \bar{\iota}_{init}^{\mathcal{B}\Rsh\hat{v}})$, where S^B denotes the states that are contained in B. Moreover, $\overline{\mathbf{P}}^{\mathcal{B}\Rsh\hat{v}}(\hat{v},\hat{v}) = \underline{\mathbf{P}}^{\mathcal{B}\Rsh\hat{v}}(\hat{v},\hat{v}) = 1$, $\overline{\mathbf{P}}^{\mathcal{B}\Rsh\hat{v}}(s,\hat{v}) = \overline{\mathbf{P}}^{\mathcal{I}}(s,v)$, $\underline{\mathbf{P}}^{\mathcal{B}\Rsh\hat{v}}(s,\hat{v}) = \underline{\mathbf{P}}^{\mathcal{I}}(s,v)$, and $\overline{\mathbf{P}}^{\mathcal{B}\Rsh\hat{v}}(s,v) = \underline{\mathbf{P}}^{\mathcal{B}\Rsh\hat{v}}(s,v) = 0$ for all $s \in S^{\mathcal{D}|_B}$. Lastly, $\mathbf{P}^{\mathcal{B}\Rsh\hat{v}}(s,t) = \mathbf{P}(s,t)$ for all $s \in S$, $t \in S\backslash\{v\}$, and $\iota_{init}^{\mathcal{B}\Rsh\hat{v}}(v) = 1$.

Intuitively, as $\mathcal{D} \models_U \mathcal{I}$ it follows that, by construction, $\mathcal{B}\Rsh\hat{v} \models_U \mathcal{I}\Rsh\hat{v}$.

Assumption 2
In the rest of this Chapter, we assume that all IDTMCs $\mathcal{I}$ are graph preserving, meaning that there exists no lower transition probability function such that $\underline{\mathbf{P}}^{\mathcal{I}}(s,t) = 0$ for any $s,t \in S^{\mathcal{I}}$. This implies that there exists no $\mathcal{D} \models_U \mathcal{I}$ such that some $s \in S^{\mathcal{I}}$ is not reachable in $\mathcal{D}$.

Given this assumption, we can prove the following theorem.

Theorem 6.4
Given an IDTMC $\mathcal{I}$, a state $s \in S^{\mathcal{I}}_{re}$, and any pair of BSCCs $B^{\mathcal{D}}, B'^{\mathcal{D}'} \subseteq S^{\mathcal{I}}$ with $B \neq B'$, it holds that $s \in B$ implies $s \notin B'$.

In other words, every state $S^{\mathcal{I}}_{re}$ corresponds to precisely one BSCC in $\mathcal{I}$.

Proof 21
Assume that for some IDTMC $\mathcal{I}$ there exists a state s, two BSCCs $B^{\mathcal{D}}, B'^{\mathcal{D}'}$ with $B \not= B'^{\mathcal{D}'}$, and $\mathcal{D}, \mathcal{D}' \models \mathcal{I}$, such that $s \in B^{\mathcal{D}}$ and $s \in B'^{\mathcal{D}'}$. As $B^{\mathcal{D}} \neq B'^{\mathcal{D}'}$, there exists at least one state $t \in S^{\mathcal{I}}$ that is only contained in either $B^{\mathcal{D}}$ or $B'^{\mathcal{D}'}$, i.e., $t \in (B\backslash B') \cup (B'\backslash B)$.

Without loss of generality, let us assume $t \in B^{\mathcal{D}}$. Hence, by Def. 3.9, there exists a path from s to t in $\mathcal{D}$. However, as noted the graph relations for all satisfying DTMCs of graph preserving IDTMCs are identical. Thus, there must also exist a path from s to t in $\mathcal{D}'$. However, $t \notin B'^{\mathcal{D}'}$, which contradicts Def. 3.9. □

Thus, we can conclude for graph preserving IDTMCs that any state $s \in S^{\mathcal{I}}$ belongs only to a singular BSCC B, i.e., the BSCCs for all satisfying DTMCs are identical. This allows us, to optimize the reachability of B ($Pr^{\mathcal{D}}(\Diamond B)$) and the stationary distribution restricted to B ($\theta^{\mathcal{D}|_B}$) of Eq. (6.1) independently as shown in the following.

Therefore, we can intuitively calculate the stationary distributions for a state $s \in \mathcal{I}$ using the following theorem.

Theorem 6.5
Given an IDTMC $\mathcal{I}$, a state s, and the corresponding BSCC $B^{\mathcal{I}}$ with $s \in B^{\mathcal{I}}$, we can calculate the minimum and maximum stationary distribution using

$$\underline{\theta}^{\mathcal{I}}(s) = \underline{Pr}^{\mathcal{I}}(\Diamond B) \cdot \frac{\underline{\mathbb{E}}^{\mathcal{I}\restriction^{\hat{v}}}[vt_s]}{\underline{\mathbb{E}}^{\mathcal{I}\restriction^{\hat{v}}}[vt_s] + \sum_{t \in S \setminus \{s\}} \mathbb{E}^{\mathcal{B}\restriction^{\hat{v}}}[vt_t]}$$

and

$$\overline{\theta}^{\mathcal{I}}(s) = \overline{Pr}^{\mathcal{I}}(\Diamond B) \cdot \frac{\overline{\mathbb{E}}^{\mathcal{I}\restriction^{\hat{v}}}[vt_s]}{\overline{\mathbb{E}}^{\mathcal{I}\restriction^{\hat{v}}}[vt_s] + \sum_{t \in S \setminus \{s\}} \mathbb{E}^{\mathcal{B}\restriction^{\hat{v}}}[vt_t]}$$

similar to Eq. (6.1) using Eq. (6.2).

We provide a short proof by contradiction of this theorem

Proof 22
Let us assume that there exists a DTMC $\mathcal{D} \models_U \mathcal{I}$ such that $\overline{\theta}^{\mathcal{I}}(s) < \theta^{\mathcal{D}}(s)$. Using Theorem 6.5 and Eq. (6.1), we can thus conclude that either

$$Pr^{\mathcal{D}}(\Diamond B) > \overline{Pr}^{\mathcal{I}}(\Diamond B) \qquad \text{or} \qquad \frac{\mathbb{E}^{\mathcal{B}\restriction^{\hat{v}}}[vt_s]}{\sum_{t \in S} \mathbb{E}^{\mathcal{B}\restriction^{\hat{v}}}[vt_t]} > \frac{\overline{\mathbb{E}}^{\mathcal{I}\restriction^{\hat{v}}}[vt_s]}{\overline{\mathbb{E}}^{\mathcal{I}\restriction^{\hat{v}}}[vt_s] + \sum_{t \in S \setminus \{s\}} \mathbb{E}^{\mathcal{B}\restriction^{\hat{v}}}[vt_t]}\,.$$

However, as $\mathcal{D} \models_U \mathcal{I}$ and $\mathcal{B}\restriction^{\hat{v}} \models_U \mathcal{I}\restriction^{\hat{v}}$, by the definition of $\overline{Pr}^{\mathcal{I}}$ and $\overline{\mathbb{E}}^{\mathcal{I}\restriction^{\hat{v}}}$ this leads to a contradiciton. The proof for $\underline{\theta}^{\mathcal{I}}(s)$ follows analogously by inverting the inequalities. □

As s belongs to a singular BSCC B, $\mathcal{B}\restriction^{\hat{v}}$ is necessarily constructed from B.

Vllt muss nochmal anders: zweiter term nicht notwendiger Weise maximal für BPfeil. Muss ein BPFeil aus für dieses summe optimieren

Chapter 7

Conclusion

Appendix A

Appendix

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